BIO 004 · HUMAN ANATOMY · SOLANO COMMUNITY COLLEGE

Learning Workbook.

Class 1 · Mon / Wed Afternoon · CRN 80650

PART A

How this works

What the book is for, what we each commit to, how to find anything on the course site, and how a brain dump runs.

PART B

The seventeen weeks

One page set per week. What to do before class, what happens in class, the competency loop, the lab structures, and three lines about your own learning.

PART C

Practice

The brain dump practice bank and blank practice pages, for use all term.

Solano Community College · Fall 2026

Dr. Sharilyn Rennie

Contents

PART A

How this workbook works

Your workbook tells you what to do. Your course site gives you the resources to do it.

This workbook is not a second set of notes and it is not a textbook. Everything you need to learn from lives on the course site: the concept videos, the notes packets, the pre-work sheets, the Atlas, the Loops, Mastery OS, the lab sprints and the recall cards. This book is the roadmap through all of it. It tells you what to do, gives you room to retrieve and organize what you know, and lets you watch your own learning change over the term.

Bring it to every class.

Every class day opens with a brain dump on paper, and the paper is this book. If you leave it at home you have left the graded work at home.

Work it in order, one week at a time.

Each week follows the same shape, so after two or three weeks you will stop reading the instructions and just work. Before the class day you prepare. In class you retrieve, rebuild and apply. After class you space it out. At the end of the week you write three lines about what changed.

The back of the book is yours to use all term.

The brain dump practice bank is a menu of every content prompt in the course. The practice pages behind it are blank on purpose. Pick a prompt, set a clock, produce what you can, then check yourself against your own materials.

THE CONTRACT

What you can expect from me

I will:

- Clearly identify what you are responsible for learning.
- Provide organized, accessible resources for learning it.
- Teach and clarify difficult concepts.
- Give you frequent opportunities to discover what you know and what you still need to learn.
- Provide meaningful practice before high-stakes assessments.
- Adjust instruction when I see that the class has a meaningful learning gap.
- Help when you are genuinely stuck and can show me what you have already attempted.
- When you have not yet made an attempt, help you re-steer toward the right course resource or learning strategy rather than simply giving you the answer.

What I expect from you

You will:

- Complete assigned preparation before class.
- Attempt retrieval before looking at notes or answers.
- Use your course resources to investigate what you do not know.
- Bring your attempts when asking for help. You do not have to be correct. You do need to have started.
- Correct mistakes rather than simply noting that an answer was wrong.
- Participate meaningfully with your team and contribute to your team's learning.
- Return to older material through spaced retrieval.
- Ask questions when you genuinely do not understand.
- Gradually become more independent in deciding what you need to practice and which tool will help you practice it.

I will work very hard to remove unnecessary friction from finding your learning. I will not remove the productive effort required to do your learning.

FIND IT

Where everything lives

Everything on the course site opens from one place. On any page, open Course tools and either type what you want in the search box or open the group it lives in. The groups are the same everywhere, so learning them once means you never hunt again.

This week

- **Today** . The one thing to do now, and your other class days.
- **Course calendar** . Every class day and what to prepare for it.
- **How to videos** . Short tours of the course. Scan a code, watch on your phone.
- **Study With Me** . Community study sessions. Hosting counts the same as attending.
- **Anatomy Games** . Taboo, Memory Match and more. Team games you play out loud.

Course materials

- **Course materials** . Notes, pre-work, videos, workbooks and decks, by module.
- **Exam modules** . Exactly what each exam covers.
- **OpenStax reference** . Free online book, for a second explanation or more figures. Not required.

Lab materials

- **Lab sprints** . Every structure you are responsible for on the models.
- **Digital Atlas** . Turn the structures around and look at them.
- **Tissue Chart Practice** . The whole chart behind reveal boxes, the same slides as lab.
- **Loops** . Image loops for fast visual practice.
- **Histology help** . Every free slide tool, sorted by the kind of help you need.

Practice and recall

- **Mastery OS** . Your cards, your weak spots and a plan around them.
- **Recall cards** . Straight into the cards that are due today.
- **Study Protocol** . A flow chart of exactly what to try this week.
- **Brain dump practice** . Spin for a prompt, set your clock, write it on paper, check yourself.
- **Draw it from memory** . Draw the structure first, then check it against the list.
- **Practice Exam Builder** . Build a fresh full-format exam any time, then track your scores.
- **Practice exam** . A 100 point paper in the real format, scored, with the reasoning.
- **Muscle charts, I O A** . Origins, insertions, actions and innervation, drilled interactively.
- **Tissue Chart Practice** . Pick the columns, flip to Recall, reveal as you go.
- **What I got done today** . And what you meant to do and did not.

About the course

- **Syllabus** . Your section's syllabus.
- **Grade calculator** . Where you stand.
- **Virtual Office** . Ask a question where the whole class sees the answer.
- **About this course** . Why the course works the way it does.
- **Need motivation?** . Short reads for hard weeks.
- **Device policy** . Sign it, then save the PDF for Canvas.
- **Course home** . Back to the front door.

The directions get shorter on purpose

You will notice these directions get shorter as the term goes on. Through Exam 1 they give you the full path. After that they give you the tool name. By the last modules they will simply say to use the right retrieval tool, because by then you will know which one that is. That is deliberate. Learning to navigate your own course is part of the course.

EVERY CLASS DAY

How a brain dump runs

A brain dump is not always a check on last night's pre-work. Some days it reaches back to material from weeks ago. Some days it asks you to connect old material to new. Some days it asks about your own learning rather than about anatomy. You will not know which kind you are getting, so the only reliable strategy is to keep everything retrievable.

Round 1, retrieve

Everything closed. No notes, no site, no phone, no AI, no teammate. Write and draw everything you can produce. This is not meant to be beautiful or complete. It is a snapshot of what your brain can hand you right now, and an incomplete snapshot is still useful information. This round is the one that gets graded, so it needs to be your own work in one colour, and it needs to be worth reading.

The check does not always run the same way

Some days I collect the dumps at the end of round 1 and you get them back marked. Some days we check in the room against your resources. Some days the check happens inside your team, where you compare what each of you produced and work out between you what is missing. You will not always know in advance which one you are getting, so put everything you have into round 1.

Whenever the check happens, use a second colour

Corrections and additions go in a different colour from your round 1 work, every time, whether you are checking against the site, against your notes or against your team. The two colours show you exactly what you did not know, which is what makes the page worth rereading before the exam. Nothing you add in the second colour changes the grade on round 1.

Sometimes, team build

On some days the resources close again and your team rebuilds the concept together on one page. Compare what each of you produced, explain your reasoning, find where you disagree and settle it. I will be circulating and listening, and what I hear decides what I teach next.

There is no answer key

There is no answer key for brain dumps, and that is on purpose. You already have the information. Finding the right page in your own materials and correcting your own understanding is the skill. If something turns out to be a real conceptual misunderstanding, bring it to me and I will teach it. What I will not do is the looking-up for you.

THE COMPETENCY LOOP

Five steps, every competency

Every competency in this course goes through the same five steps. Work across the row and tick each box as you finish it. The order is the point. Producing before you check, and checking in a second colour, is what turns reading into knowing.

Step 1
Name it
Write the competency out in your own words before you look anything up. If you cannot say what it is asking you to be able to do, that is the first thing to fix.

Step 2
Note sheet
Put it on your study box note sheet with the notes packet and the reading open beside you. This is the pre-work step, and it comes before the video.

Step 3
Video, second colour
Now watch the concept video and add to the same sheet in a different colour. The second colour is the record of what you did not already have.

Step 4
Blind recall
Close everything. Whiteboard or blank paper. Produce it from memory right away, while you still think you know it. This is where you find out whether you do.

Step 5
Memory hook
Build something that makes it stick: a mnemonic, a short story, an analogy, a drawing. Write the hook on the line so you can reuse it before the exam.

MODULE 1

Foundations, cells, tissues (histology) and the integument

TBL 1 Cells. TBL 2 Histology.

Weeks **1 to 4** Exam **1**, **Wed Sep 9** Lab practical and lecture exam **same day**

Week 1 Foundations, orientation and the cell

MODULE 1

BEFORE MONDAY, AUG 17

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Intro to Anatomy Lab, Lab Safety and Introduction to Histology
  Course tools → Lab materials → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Bring this workbook

MON AUG 17 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Intro to Anatomy, team formation and Canvas setup
- [ ] Lab: Intro to Anatomy Lab, Lab Safety and Introduction to Histology
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, AUG 19

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Epithelial Tissues
  Course tools → Lab materials → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Bring this workbook

WED AUG 19 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Guided Cell Activity
- [ ] Lab: Epithelial Tissues
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Practice and recall → Mastery OS

- [ ] Loops for the lab material
  Course tools → Lab materials → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Practice and recall → Brain dump practice

Competency loop, week 1

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Levels of structural organization
List the six levels of structural organization in order from chemical to organismal and state what each level contains.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Anatomical position
Place the body in anatomical position, define supine and prone, and explain why anatomical position is the fixed reference for every directional term.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Anatomical planes and sections
Name the sagittal, midsagittal, parasagittal, frontal, transverse, and oblique planes and state the two parts each divides the body into.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Directional terms
Use the paired directional terms precisely, including superior and inferior, anterior and posterior, medial and lateral, proximal and distal, superficial and deep, cranial and caudal, and ipsilateral and contralateral.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Regional terms
Identify the major anterior and posterior regional terms and sort each as axial or appendicular.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Body cavities
Distinguish the dorsal and ventral body cavities and name every subdivision of each, including the cranial, vertebral, thoracic, pleural, pericardial, abdominal, and pelvic cavities.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Serous membranes
Define a serous membrane and differentiate its parietal and visceral layers across the pleura, pericardium, and peritoneum.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Mediastinum
Map the boundaries and the four compartments of the mediastinum and identify a major structure contained in each.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Peritoneal relationships
Classify abdominopelvic organs as intraperitoneal or retroperitoneal using SADPUCKER and define the mesentery, omenta, and mesocolon.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Abdominopelvic regions and quadrants
Locate organs on the nine-region and four-quadrant grids and name the planes that draw each grid.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Generalized cell regions
Name the three basic regions of a generalized cell, the plasma membrane, cytoplasm, and nucleus, and state what each contains.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Plasma membrane structure
Describe the plasma membrane components, distinguish integral from peripheral proteins, and identify the surface specializations microvilli, cilia, and flagellum.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Nucleus structure
Identify the nuclear envelope, nuclear pores, nucleolus, nucleoplasm, chromatin, and chromosomes and state the role of each.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Organelle identification

Identify the membranous and non-membranous organelles and state the structural job of each, including the endomembrane system path from rough ER to exocytosis.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Embryonic germ layers

Name the three primary germ layers, ectoderm, mesoderm, and endoderm, and the tissues each one gives rise to.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Epithelial tissue identification

Classify epithelial tissue by cell layers and cell shape and identify the basement membrane, apical specializations, and glandular epithelium on a slide.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cell junctions

Identify the five epithelial cell junctions, what each does, what it links to, and its major protein.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

Something that still needs work

My priority for spaced retrieval next week

LEARNING TO LEARN · WEEK 1

This course may feel different

If you are used to sitting in a lecture, taking notes and studying those notes the night before an exam, this course is going to feel strange for a couple of weeks. The lecture is online, so class time is not for receiving information. It is for producing it.

That means you will regularly be asked to do things you cannot do yet. You will start a class by writing down everything you know about a topic and discovering that it is less than you thought. You will draw structures from memory and get them wrong. You will sit with a team and find out that someone else understood something you missed.

None of that is a sign the course is going badly. It is the course working. Research on learning is consistent on one point: the feeling of ease while you study is a poor predictor of what you will still know later. Struggling to produce something is uncomfortable and it is also what makes it stay.

So expect the first two weeks to feel awkward. Do the pre-work before class, bring your workbook, and give the retrieval a real attempt before you open anything. The awkwardness fades. What you build in its place is the ability to actually use this material.

Week 2 Cell anatomy and connective tissue

BEFORE MONDAY, AUG 24

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Connective Tissue I
  Course tools → Lab materials → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON AUG 24 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Connective Tissue I
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, AUG 26

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Connective Tissue II
  Course tools → Lab materials → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Bring this workbook

WED AUG 26 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Interactive Histology Lecture
- [ ] Lab: Connective Tissue II
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Practice and recall → Mastery OS
- [ ] Loops for the lab material
  Course tools → Lab materials → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Practice and recall → Brain dump practice

Competency loop, week 2

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Connective tissue identification

Describe connective tissue as cells, fibers, and ground substance and recognize the connective tissue proper subtypes, cartilage, bone, and blood on a slide.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

Something that still needs work

My priority for spaced retrieval next week

LEARNING TO LEARN · WEEK 2

Comfort is not the same as learning

Here is one of the most reliable findings in the study of learning, and one of the most annoying. The study methods that feel best while you are doing them are usually not the ones that work.

Rereading your notes feels good. The words are familiar, everything makes sense, and you finish feeling prepared. Highlighting feels productive. Watching a video a second time feels like studying. All of these produce a sense of fluency, and fluency feels like knowledge.

It is not. Recognizing something when it is in front of you is a much easier task than producing it when it is not. The exam asks you to produce. So does the lab practical, and so does every patient you will ever have.

The methods that work tend to feel worse. Closing the book and writing what you remember feels clumsy. Coming back to old material feels like a waste of time when new material is piling up. Mixing topics together feels disorganized.

When you catch yourself choosing the comfortable option, that is the moment to notice. Ask whether you are practicing recognition or practicing production. Only one of those is what you will be asked for.

Week 3 Histology and the integument

MODULE 1

BEFORE MONDAY, AUG 31

- ☐ Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- ☐ Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- ☐ Pre-study the lab structures for Muscle Tissue and Nervous Tissue
  Course tools → Lab materials → Lab sprints

- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON AUG 31 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Muscle Tissue and Nervous Tissue
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, SEP 2

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Integumentary
  Course tools → Lab materials → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Practice and recall → Recall cards
- [ ] Bring this workbook

WED SEP 2 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Exam 1 Review Game
- [ ] Lab: Integumentary
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Practice and recall → Mastery OS
- [ ] Loops for the lab material
  Course tools → Lab materials → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Practice and recall → Brain dump practice

Competency loop, week 3

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Body membranes

Distinguish the cutaneous, mucous, serous, and synovial membranes by composition and location.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Skin layers and hypodermis

Name the epidermis and dermis and the hypodermis beneath them and identify the tissue that composes each.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Epidermal strata and cells
List the five epidermal strata in order deep to superficial and identify the keratinocytes, melanocytes, tactile cells, and dendritic cells.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Dermis layers
Distinguish the papillary and reticular layers of the dermis by tissue and identify the structures and sensory corpuscles each contains.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Accessory structures of the skin
Identify the accessory structures of the skin including the parts of a hair and follicle, the parts of a nail, and the five gland types and their secretion modes.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

Something that still needs work

My priority for spaced retrieval next week

LEARNING TO LEARN · WEEK 3

Why I make you try before looking

You will notice a pattern in how this course is built. You work the pre-work sheet before you watch the video. You do the brain dump before you open your notes. You draw the structure before you see the labelled image.

This is not to make things harder for the sake of it. Attempting something before you are taught it changes what happens when you are taught it. When you have already tried, you know where your gaps are, so the explanation lands on a question you actually have instead of washing over you. Cognitive psychologists have studied this under a few names, including problem solving before instruction and the pretesting effect, and the pattern holds even when the first attempt is mostly wrong.

That last part matters. The attempt does not have to succeed to be worth making. A wrong answer that you generated yourself prepares you to learn the right one better than no answer at all.

So when the sheet asks you something you have not been taught yet, write your best guess. When the brain dump asks for something you half remember, write the half. Then watch the video, and notice how much more of it you can actually use.

Week 4 Exam 1, then bone tissue

MODULES 1 AND 2

Week 4 sits in two modules. Exam 1 closes Module 1 this week, and the teaching that starts Module 2 begins in the same week. The full week page, with the checklists and the competency loop, is printed at the front of Module 2. Work it there.

Lab structures Module 1

EXAM 1 PRACTICAL

Tick the block name once you can identify it on the slide or the model, fill in function and location where the lines are there, then tick each structure you can point to. Anything not on this list is context, not something you will be asked to identify. The reference tables, regional and directional terms, quadrants and the rest, are in the standalone Module 1 structure list on the course site.

Epithelial Tissues

☐ General structures, on every epithelial slide

☐ Apical surface ☐ Lateral surface ☐ Basal surface ☐ Basement membrane ☐ Lamina propria, when present ☐ Nuclei
☐ Lumen or free surface ☐ Cilia, when present ☐ Microvilli, when present ☐ Goblet cells, when present
☐ Keratin, when present

☐ Simple squamous

FUNCTION ______________________ LOCATION ______________________

☐ Keratinized stratified squamous

FUNCTION ______________________ LOCATION ______________________

☐ Non-keratinized stratified squamous

FUNCTION ______________________ LOCATION ______________________

☐ Simple cuboidal

FUNCTION ______________________ LOCATION ______________________

☐ Stratified cuboidal

FUNCTION ______________________ LOCATION ______________________

☐ Simple columnar

FUNCTION ______________________ LOCATION ______________________

☐ Pseudostratified ciliated columnar

FUNCTION ______________________ LOCATION ______________________

☐ Transitional

FUNCTION ______________________ LOCATION ______________________

Connective Tissues

☐ Loose areolar

FUNCTION ______________________ LOCATION ______________________

☐ Collagen fibers ☐ Elastic fibers ☐ Fibroblasts ☐ Mast cells

☐ Adipose

FUNCTION ______________________ LOCATION ______________________

☐ Adipocytes ☐ Fat vacuoles containing lipids ☐ Blood vessels

☐ Reticular

FUNCTION ______________________ LOCATION ______________________

☐ Reticular fibers

- [ ] **Dense regular**

FUNCTION ______________________ LOCATION ______________________

- [ ] Collagen fibers
- [ ] Fibroblasts

- [ ] **Dense irregular**

FUNCTION ______________________ LOCATION ______________________

- [ ] Collagen fibers
- [ ] Fibroblasts

- [ ] **Elastic**

FUNCTION ______________________ LOCATION ______________________

- [ ] Elastin fibers

- [ ] **Hyaline cartilage**

FUNCTION ______________________ LOCATION ______________________

- [ ] Perichondrium
- [ ] Lacunae
- [ ] Chondrocytes
- [ ] Extracellular matrix

- [ ] **Elastic cartilage**

FUNCTION ______________________ LOCATION ______________________

- [ ] Perichondrium
- [ ] Lacunae
- [ ] Chondrocytes
- [ ] Extracellular matrix
- [ ] Elastin fibers

- [ ] **Fibrocartilage**

FUNCTION ______________________ LOCATION ______________________

- [ ] Lacunae
- [ ] Chondrocytes
- [ ] Extracellular matrix
- [ ] Collagen fibers
- [ ] No perichondrium

- [ ] **Osseous tissue**

FUNCTION ______________________ LOCATION ______________________

- [ ] Osteons
- [ ] Central (Haversian) canal
- [ ] Lacunae
- [ ] Lamellae
- [ ] Osteocytes
- [ ] Canaliculi

Muscle Tissues

- [ ] **Skeletal muscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Striations, dark and light
- [ ] Nuclei, multiple and peripheral
- [ ] Sarcolemma

- [ ] **Cardiac muscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Branching fibers
- [ ] Intercalated discs
- [ ] Single central nucleus
- [ ] Striations

- [ ] **Smooth muscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Spindle-shaped cells
- [ ] Single central nucleus
- [ ] No striations

Nerve Tissues

- [ ] **Multipolar neuron**

FUNCTION ______________________ LOCATION ______________________

- [ ] Cell body (soma)
- [ ] Nucleus
- [ ] Nucleolus
- [ ] Dendrites
- [ ] Axon
- [ ] Axon hillock
- [ ] Synaptic bulb

- [ ] **Spinal cord**

FUNCTION ______________________ LOCATION ______________________

- [ ] White matter
- [ ] Grey matter
- [ ] Ventral horn
- [ ] Dorsal horn
- [ ] Lateral horn, when present

- [ ] **Cerebral cortex**

FUNCTION ______________________ LOCATION ______________________

- [ ] Pyramidal cells

- [ ] **Cerebellum**

FUNCTION ______________________ LOCATION ______________________

- [ ] Purkinje cells

- [ ] **Nerve, cross section**

FUNCTION ______________________ LOCATION ______________________

- [ ] Axon - [ ] Schwann cells - [ ] Nodes of Ranvier - [ ] Epineurium - [ ] Perineurium - [ ] Endoneurium

Blood Vessels and Formed Elements

- [ ] **Artery**

FUNCTION ______________________ LOCATION ______________________

- [ ] Tunica intima - [ ] Endothelium - [ ] Internal elastic lamina - [ ] Tunica media - [ ] Smooth muscle
- [ ] Dense elastic connective tissue - [ ] Tunica externa - [ ] Lumen - [ ] Erythrocytes

- [ ] **Vein**

FUNCTION ______________________ LOCATION ______________________

- [ ] Tunica intima - [ ] Endothelium - [ ] Tunica media with smooth muscle - [ ] Tunica externa - [ ] Lumen - [ ] Erythrocytes

- [ ] **Neutrophil**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Basophil**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Eosinophil**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Monocyte**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Lymphocyte**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Erythrocyte**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Thrombocyte**

FUNCTION ______________________ LOCATION ______________________

Integumentary System

- [ ] **Stratum corneum**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Stratum lucidum**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Stratum granulosum**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Stratum spinosum**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Stratum basale**

FUNCTION ______________________ LOCATION ______________________

- [ ] **Dermis, on slides and models**

- [ ] Papillary layer
- [ ] Reticular layer
- [ ] Areolar connective tissue
- [ ] Dense irregular connective tissue
- [ ] Dermal papillae
- [ ] Eccrine sweat gland and duct
- [ ] Hair
- [ ] Hair follicle
- [ ] Hair root
- [ ] Hair bulb
- [ ] Hair shaft
- [ ] Arrector pili muscle
- [ ] Sebaceous gland and duct
- [ ] Apocrine gland and duct
- [ ] Meissner's corpuscles
- [ ] Free nerve endings
- [ ] Capillary loops

- [ ] **Hypodermis**

- [ ] Adipose connective tissue
- [ ] Areolar connective tissue
- [ ] Lamellated (Pacinian) corpuscle

MODULE 2

Skeletal system: bone tissue, axial and appendicular skeleton, joints

TBL 3 Bone tissue, cartilage and microanatomy.

Weeks **4 to 7** Exam **2**, **Wed Sep 30** Lab practical and lecture exam **same day**

Week 4 Exam 1, then bone tissue

MODULES 1 AND 2

MON SEP 7 · CAMPUS CLOSED

BEFORE WEDNESDAY, SEP 9

- [ ] Run a gap check on both banks and work the three day plan it builds
  Course tools → Practice and recall → Mastery OS
- [ ] Produce five brain dump prompts from this module on blank paper
  Course tools → Practice and recall → Brain dump practice
- [ ] Walk the lab structure list for this module and tick what you can name blind
  Course tools → Lab materials → Lab sprints
- [ ] Sit a practice exam before exam day
  Course tools → Practice and recall → Practice Exam Builder

WED SEP 9 · EXAM DAY

- [ ] Lab practical
- [ ] Lecture exam

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Practice and recall → Mastery OS
- [ ] Loops for the lab material
  Course tools → Lab materials → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Practice and recall → Brain dump practice

Competency loop, week 4

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Cartilage types and structure
Identify the three cartilage types by fiber content and location and label chondrocytes, lacunae, matrix, and perichondrium.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Cartilage growth modes
Distinguish appositional from interstitial cartilage growth by where each begins and the direction of expansion.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Bone classification by shape
Classify a bone as long, short, flat, irregular, or sesamoid and give an example of each.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Gross anatomy of a long bone
Label the diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and marrow regions on a long bone.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________

Compact and spongy bone microanatomy
Label the parts of an osteon and contrast compact bone with the trabeculae of spongy bone by structure and location.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________

The four bone cells
Identify the osteogenic cell, osteoblast, osteocyte, and osteoclast and state the role and origin of each.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________

Ossification and growth plate zones
Contrast intramembranous and endochondral ossification and list the five growth plate zones in order from epiphysis to diaphysis.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 4

Forgetting is why we come back

You are going to forget things in this course. All of it, some of it, repeatedly. That is not a defect in your studying and it is not a sign that you are not cut out for this.

Forgetting is normal and it is also useful. Every time you let something slip a little and then pull it back, the pulling back does more for you than the original learning did. This is why the Loops keep returning to material you covered weeks ago, and why some brain dumps ask about a topic you have not thought about since Module 1.

The mistake is treating a topic as finished. You learned the tissues, you passed the exam question, you moved on. Six weeks later the same tissue shows up on a slide inside an organ and you cannot name it. Nothing went wrong. You simply never came back.

Build the return into your week rather than hoping it happens. Recall cards do this automatically. So does picking two brain dump prompts from an old module before you start on the new one. Ten minutes on old material is worth more than another hour on the material you just covered.

Week 5 Bone microanatomy and the skull

MODULE 2

BEFORE MONDAY, SEP 14

☐ Work the pre-work sheet with your notes packet and the reading open beside you

Course tools → Course materials

- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Bone Tissue, Long Bone and Microanatomy
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

MON SEP 14 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Exam 1 rebuttals + Long Bone and Bone Microanatomy
- [ ] Lab: Bone Tissue, Long Bone and Microanatomy
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, SEP 16

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Axial Skeleton: Skull
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

WED SEP 16 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Skull Walk-Through
- [ ] Lab: Axial Skeleton: Skull
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 5

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Cranial and facial bones

Distinguish the eight cranial bones from the fourteen facial bones and name the region each bone forms.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________________

Significant skull bone markings

Identify the key markings on each cranial and facial bone, including features of the sphenoid, ethmoid, temporal, occipital, maxillae, and mandible.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ________________

Sutures and fontanelles

Locate the four major sutures and the pterion by the bones they join and name the four fontanelles and where each lies.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Orbit, nasal cavity, palate, and sinuses

Identify the bones forming the orbit, nasal septum, and hard palate and locate the four paranasal sinuses by the bone each occupies.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Major skull foramina and contents

Name the major skull foramina by the bone each pierces and state the nerve or vessel that passes through each.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Vertebral column regions and curvatures

Name the five regions of the vertebral column with the vertebra count of each and identify the four curvatures as primary or secondary.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Typical vertebra and intervertebral disc

Label the parts of a typical vertebra and the anulus fibrosus and nucleus pulposus of an intervertebral disc.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Regional and specialized vertebrae

Distinguish cervical, thoracic, and lumbar vertebrae by an identifying feature and describe the atlas, axis, dens, sacrum, and coccyx.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Sternum and ribs

Identify the parts of the sternum and rib markings and classify the twelve rib pairs as true, false, or floating by anterior attachment.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 5

What getting something wrong can teach you

Most students treat a wrong answer as a verdict. You missed it, you feel bad, you move on. That wastes the most useful piece of information you get all week.

A wrong answer tells you what kind of problem you have, and there are three common kinds. Sometimes you could not recall it at all, which means the material never got encoded well enough or you have not come back to it. Sometimes you recalled it, but wrong, which usually means it is tangled with something similar. Sometimes you knew the fact perfectly and still could not apply it to the question in front of you, which means you learned it as a label instead of as an idea.

Each of those needs a different fix. The first needs more retrieval practice. The second needs you to sit down with the two things you are confusing and write out what separates them. The third needs application questions and cases, not more flashcards.

So when you check your brain dump this week, do not just mark what was wrong. Write one word beside it: recall, confused or apply. Then choose the fix that matches.

Week 6 Spine, thoracic cage and the upper limb

MODULE 2

BEFORE MONDAY, SEP 21

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Axial Skeleton: Vertebrae and Ribs
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

MON SEP 21 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Spine Walk-Through
- [ ] Lab: Axial Skeleton: Vertebrae and Ribs
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, SEP 23

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Upper Extremity
  Course tools → Lab sprints
- [ ] Run the recall cards that are due

Course tools → Recall cards

- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

WED SEP 23 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Upper Extremity
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 6

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Pectoral girdle and its markings Identify the markings of the clavicle and scapula and explain how the clavicle links the upper limb to the axial skeleton.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Humerus, radius, and ulna markings Identify the key markings of the humerus, radius, and ulna and state which forearm bone is lateral and which forms the elbow hinge.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Bones of the hand Name the eight carpals in their two rows and the metacarpals and phalanges and locate the carpal tunnel and its contents.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Bone markings vocabulary Classify a named bone marking as a projection, depression, or opening and match each term such as process, tuberosity, fossa, or sulcus to its description.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Hip bone and pelvis Name the three bones that fuse at the acetabulum, identify the markings of the ilium, ischium, and pubis, and describe the pelvis and its sex differences.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Femur and patella markings Identify the key markings of the femur and the base and apex of the patella.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Tibia and fibula markings Identify the markings of the tibia and fibula and state which bone bears weight and how to tell them apart.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Bones and arches of the foot Name the seven tarsals, the metatarsals, and the phalanges and identify the three arches of the foot.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Structural and functional joint classification Classify a joint by structure as fibrous, cartilaginous, or synovial and by function as synarthrosis, amphiarthrosis, or diarthrosis and state the criteria for each.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Fibrous and cartilaginous joints
Describe the three fibrous joint types and the two cartilaginous joint types by structure, mobility, and example.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Synovial joint structure and accessory parts
Identify the structures of a synovial joint, its accessory structures such as menisci, labra, bursae, and tendon sheaths, and its nerve and blood supply.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Movements at synovial joints
Demonstrate and name the gliding, angular, rotational, and special movements permitted at synovial joints.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Six synovial joint types
Name the six synovial joint types with their axes of movement and give a body example of each.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Shoulder and knee joint structure
Describe the articulating bones and supporting ligaments of the glenohumeral and tibiofemoral joints and identify the rotator cuff, menisci, and cruciate and collateral ligaments.
☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 6

Your team is a learning tool

Your team is not a social arrangement and it is not a way to divide the work. It is one of the most efficient diagnostic instruments in the room.

When you explain something to a teammate, you find out immediately whether you understand it or only recognize it. Words come out fluently for things you know and stall for things you do not. That stall is information, and you would never get it studying alone.

When a teammate explains something differently than you would, you get a second route into the same idea. When two of you disagree about a classification, one of you is about to learn something, and often you both are, because working out why you disagree usually surfaces a distinction neither of you had clean.

This only works if you actually produce. A team where one person talks and the rest nod is a team where one person is learning. Bring your own attempt to the table, say what you think before you hear what everyone else thinks, and be willing to be the one who is wrong out loud.

The teams are permanent for a reason. Groups that stay together get past politeness and start arguing about content, which is where the learning is.

Week 7 Lower limb, joints and Exam 2

MODULES 2 AND 3

Week 7 sits in two modules. Exam 2 closes Module 2 this week, and the teaching that starts Module 3 begins in the same week. The full week page, with the checklists and the competency loop, is printed at the front of Module 3. Work it there.

Lab structures Module 2

EXAM 2 PRACTICAL

Tick the block name once you can identify it on the slide or the model, fill in function and location where the lines are there, then tick each structure you can point to. Anything not on this list is context, not something you will be asked to identify. The reference tables, regional and directional terms, quadrants and the rest, are in the standalone Module 2 structure list on the course site.

Bone Tissue and the Long Bone

☐ Osseous tissue, on the bone tissue model

☐ Osteon ☐ Central (Haversian) canal ☐ Volkmann's canal ☐ Canaliculi ☐ Lacunae ☐ Osteocytes
☐ Inner circumferential lamellae ☐ Outer circumferential lamellae ☐ Interstitial lamellae ☐ Concentric lamellae
☐ Medullary cavity ☐ Periosteum, inner osteogenic layer ☐ Periosteum, outer fibrous layer ☐ Sharpey's fibers

☐ Gross anatomy of a long bone

☐ Epiphysis ☐ Metaphysis ☐ Diaphysis ☐ Medullary cavity ☐ Endosteum ☐ Periosteum, osteogenic layer
☐ Periosteum, fibrous layer ☐ Articular cartilage ☐ Epiphyseal line, in adults ☐ Epiphyseal plate, in children
☐ Compact bone ☐ Spongy bone ☐ Collagen fibers between the lamellae ☐ Red marrow ☐ Yellow marrow

Axial Skeleton: The Skull

☐ Cranial and facial bones

☐ Frontal ☐ Occipital ☐ Parietal ☐ Temporal ☐ Sphenoid ☐ Ethmoid ☐ Maxilla ☐ Zygomatic ☐ Nasal
☐ Lacrimal ☐ Mandible

☐ Sutures

☐ Coronal ☐ Sagittal ☐ Squamosal ☐ Lambdoid

☐ Lateral skull

☐ Glabella ☐ Supraorbital foramen or notch ☐ Supraorbital margin ☐ Zygomatic arch
☐ Zygomatic process of the temporal bone ☐ Temporal process of the zygomatic bone ☐ Mastoid process
☐ External acoustic (auditory) meatus ☐ Styloid process ☐ Articular tubercle ☐ Alveolar process ☐ Anterior nasal spine
☐ Infraorbital foramen

☐ Anterior skull

☐ Supraorbital foramen or notch ☐ Supraorbital margin ☐ Glabella ☐ Perpendicular plate of the ethmoid ☐ Vomer
☐ Middle nasal concha ☐ Inferior nasal concha ☐ Middle nasal meatus ☐ Inferior nasal meatus ☐ Anterior nasal spine
☐ Alveolar process ☐ Infraorbital foramen ☐ Nasion

- [ ] **Posterior skull**

- [ ] Superior nuchal line - [ ] Inferior nuchal line - [ ] External occipital protuberance - [ ] Vomer - [ ] Occipital condyles - [ ] Palatine bone - [ ] Palatine process of the maxilla - [ ] Incisive fossa and foramen - [ ] Mastoid process

- [ ] **Base of the skull, exterior**

- [ ] Incisive foramen - [ ] Transverse palatine suture - [ ] Median palatine suture - [ ] Palatine bone - [ ] Vomer - [ ] Medial pterygoid plate - [ ] Lateral pterygoid plate - [ ] Foramen ovale - [ ] Foramen spinosum - [ ] Foramen lacerum - [ ] Foramen rotundum - [ ] Carotid canal - [ ] Jugular foramen - [ ] Hypoglossal canal - [ ] Foramen magnum - [ ] Inferior nuchal line - [ ] Superior nuchal line - [ ] External occipital protuberance - [ ] Mastoid process - [ ] Articular tubercle - [ ] Occipital condyles - [ ] Styloid process - [ ] Mandibular fossa - [ ] Zygomatic arch - [ ] Inferior orbital fissure - [ ] Choana

- [ ] **Base of the skull, interior**

- [ ] Anterior cranial fossa - [ ] Middle cranial fossa - [ ] Posterior cranial fossa - [ ] Frontal crest - [ ] Frontal sinus - [ ] Cribriform plate of the ethmoid - [ ] Crista galli - [ ] Olfactory foramina - [ ] Lesser wing of the sphenoid - [ ] Greater wing of the sphenoid - [ ] Hypophyseal fossa (sella turcica) - [ ] Dorsum sellae - [ ] Tuberculum sellae - [ ] Internal acoustic (auditory) meatus - [ ] Jugular foramen - [ ] Internal occipital protuberance - [ ] Hypoglossal canal - [ ] Foramen lacerum - [ ] Foramen magnum - [ ] Foramen spinosum - [ ] Foramen ovale - [ ] Optic canal - [ ] Anterior clinoid processes

- [ ] **Ethmoid bone**

- [ ] Crista galli - [ ] Perpendicular plate - [ ] Cribriform plate - [ ] Olfactory foramina

- [ ] **Sphenoid bone**

- [ ] Medial pterygoid plate - [ ] Lateral pterygoid plate - [ ] Hypophyseal fossa (sella turcica) - [ ] Dorsum sellae - [ ] Tuberculum sellae - [ ] Greater wing - [ ] Lesser wing - [ ] Superior orbital fissure

- [ ] **Orbit**

- [ ] Sphenoid bone - [ ] Lacrimal bone - [ ] Lacrimal canal - [ ] Optic canal (foramen) - [ ] Superior orbital fissure - [ ] Inferior orbital fissure

- [ ] **Mandible**

- [ ] Condylar process (condyle) - [ ] Coronoid process - [ ] Mandibular notch - [ ] Mental protuberance - [ ] Mental foramen - [ ] Alveolar process - [ ] Alveolar margin - [ ] Body - [ ] Ramus - [ ] Mandibular foramen - [ ] Angle of the mandible

- [ ] **Hyoid bone**

- [ ] Body - [ ] Greater cornu (greater horn) - [ ] Lesser cornu (lesser horn)

- [ ] **Nasal septum**

- [ ] Septal cartilage - [ ] Perpendicular plate of the ethmoid - [ ] Vomer

- [ ] **Fetal skull**

- [ ] Anterior fontanel - [ ] Posterior fontanel - [ ] Right and left anterolateral fontanel - [ ] Right and left posterolateral fontanel

Axial Skeleton: Spine and Thoracic Cage

- [ ] **Spinal segments**

- [ ] Cervical spine, C1 to C7 - [ ] Thoracic spine, T1 to T12 - [ ] Lumbar spine, L1 to L5 - [ ] Sacrum, S1 to S5 fused - [ ] Coccyx, fused

- [ ] **Normal spinal curves**

- [ ] Cervical - [ ] Thoracic - [ ] Lumbar - [ ] Sacrococcygeal

- [ ] **Abnormal spinal curves**

- [ ] Scoliosis - [ ] Lordosis - [ ] Kyphosis

- [ ] **C1, the atlas**

- [ ] Anterior tubercle - [ ] Posterior tubercle - [ ] Facet for the dens - [ ] Anterior arch - [ ] Posterior arch
- [ ] Superior articular facet and process - [ ] Inferior articular facet and process - [ ] Transverse process - [ ] Transverse foramen
- [ ] Vertebral foramen

- [ ] **C2, the axis**

- [ ] Dens - [ ] Superior articular facet and process - [ ] Inferior articular facet and process - [ ] Body - [ ] Transverse process
- [ ] Transverse foramen - [ ] Vertebral foramen - [ ] Spinous process - [ ] Vertebral arch

- [ ] **Typical cervical vertebra**

- [ ] Vertebral foramen - [ ] Transverse process - [ ] Transverse foramen - [ ] Body - [ ] Pedicle - [ ] Lamina
- [ ] Spinous process - [ ] Vertebral arch - [ ] Superior articular facet and process - [ ] Inferior articular facet and process

- [ ] **Typical thoracic and lumbar vertebra**

- [ ] Body - [ ] Superior articular facet and process - [ ] Inferior articular facet and process - [ ] Costal facet, thoracic only
- [ ] Transverse process - [ ] Spinous process - [ ] Lamina - [ ] Pedicle - [ ] Vertebral arch - [ ] Vertebral foramen

- [ ] **Placing a vertebra on the articulated column**

- [ ] **Rib**

- [ ] Head - [ ] Neck - [ ] Shaft

- [ ] **Sacrum and coccyx**

- [ ] Sacral promontory - [ ] Superior articular process and facet - [ ] Anterior sacral foramina - [ ] Coccyx
- [ ] Posterior sacral foramina - [ ] Sacral hiatus - [ ] Median sacral crest - [ ] Lateral sacral crest - [ ] Articular surface
- [ ] Base of the sacrum - [ ] Apex of the sacrum - [ ] Sacral tuberosity

- [ ] **Sternum**

- [ ] Manubrium - [ ] Sternal angle - [ ] Body - [ ] Xiphoid process - [ ] Jugular notch - [ ] Clavicular notch - [ ] Costal notch

- [ ] **Intervertebral disc**

- [ ] Annulus fibrosus - [ ] Nucleus pulposus

Appendicular Skeleton: Pectoral Girdle and Upper Limb

- [ ] **Clavicle**

- [ ] Sternal end - [ ] Acromial end - [ ] Conoid tubercle

- [ ] **Scapula**

- [ ] Acromion - [ ] Coracoid process - [ ] Spine of the scapula - [ ] Infraspinous fossa - [ ] Supraspinous fossa
- [ ] Subscapular fossa - [ ] Glenoid cavity - [ ] Supraglenoid tubercle - [ ] Infraglenoid tubercle - [ ] Superior border
- [ ] Superior angle - [ ] Suprascapular notch - [ ] Axillary (lateral) border - [ ] Vertebral (medial) border - [ ] Inferior angle

- [ ] **Humerus**

- [ ] Head - [ ] Anatomical neck - [ ] Surgical neck - [ ] Greater tubercle - [ ] Lesser tubercle
- [ ] Intertubercular sulcus (bicipital groove) - [ ] Deltoid tuberosity - [ ] Radial groove - [ ] Lateral epicondyle
- [ ] Lateral supracondylar ridge - [ ] Medial epicondyle - [ ] Medial supracondylar ridge - [ ] Trochlea - [ ] Coronoid fossa
- [ ] Capitulum - [ ] Radial fossa - [ ] Olecranon fossa

- [ ] **Radius**

- [ ] Head - [ ] Neck - [ ] Styloid process - [ ] Radial tuberosity - [ ] Ulnar notch

- [ ] **Ulna**

- [ ] Head - [ ] Styloid process - [ ] Ulnar tuberosity - [ ] Radial notch - [ ] Coronoid process - [ ] Trochlear notch
- [ ] Olecranon process

- [ ] **Carpals, on an articulated wrist**

- [ ] Scaphoid - [ ] Lunate - [ ] Triquetrum (triangular) - [ ] Pisiform - [ ] Hamate - [ ] Capitate - [ ] Trapezoid - [ ] Trapezium

- [ ] **Metacarpals and phalanges**

- [ ] Metacarpals 1 to 5, lateral to medial - [ ] Proximal phalanges - [ ] Middle phalanges - [ ] Distal phalanges

Appendicular Skeleton: Pelvic Girdle and Lower Limb

- [ ] **Coxal bone, the ilium region**

- [ ] Iliac crest - [ ] Auricular surface - [ ] Iliac fossa - [ ] Greater sciatic notch - [ ] Lesser sciatic notch - [ ] Ala
- [ ] Anterior superior iliac spine (ASIS) - [ ] Anterior inferior iliac spine (AIIS) - [ ] Posterior superior iliac spine (PSIS)
- [ ] Posterior inferior iliac spine (PIIS)

- [ ] **Ischium**

- [ ] Ischial spine - [ ] Ischial tuberosity - [ ] Lesser sciatic notch - [ ] Ramus of the ischium

- [ ] **Pubis**

- [ ] Pubic body - [ ] Superior ramus - [ ] Inferior ramus - [ ] Pubic crest - [ ] Pubic tubercle

- [ ] **Landmarks and joints of the pelvis**

- [ ] Acetabulum - [ ] Obturator foramen - [ ] Sacroiliac joint - [ ] Pubic symphysis

- [ ] **Femur**

- [ ] Head - [ ] Fovea capitis - [ ] Neck - [ ] Greater trochanter - [ ] Lesser trochanter - [ ] Intertrochanteric line
- [ ] Intertrochanteric crest - [ ] Gluteal tuberosity - [ ] Linea aspera, lateral lip - [ ] Linea aspera, medial lip
- [ ] Lateral supracondylar line - [ ] Medial supracondylar line - [ ] Adductor tubercle - [ ] Lateral epicondyle
- [ ] Medial epicondyle - [ ] Lateral condyle - [ ] Medial condyle - [ ] Patellar surface - [ ] Intercondylar fossa

- [ ] **Patella**

- [ ] **Tibia**

- [ ] Lateral condyle - [ ] Medial condyle - [ ] Intercondylar eminence - [ ] Tibial tuberosity - [ ] Fibular notch
- [ ] Anterior crest (border) - [ ] Medial malleolus

- [ ] **Fibula**

- [ ] Head of the fibula - [ ] Lateral malleolus

- [ ] **Tarsals**

- [ ] Talus - [ ] Calcaneus - [ ] Navicular - [ ] Medial cuneiform - [ ] Intermediate cuneiform - [ ] Lateral cuneiform - [ ] Cuboid

- [ ] **Metatarsals and phalanges**

- [ ] Metatarsals I to V, medial to lateral - [ ] Proximal phalanges - [ ] Middle phalanges - [ ] Distal phalanges

Joints

- [ ] **For any joint you are given, be ready to**

- [ ] Classify it as fibrous, cartilaginous, or synovial.
- [ ] Name the type of synovial joint, for example hinge, ball and socket, saddle.
- [ ] Classify it by movement: synarthrosis (immovable), amphiarthrosis (slightly movable), diarthrosis (freely movable).
- [ ] Name the specific movements that occur there, if any. Flexion, extension, internal rotation, and so on.

- [ ] **Named joints, know these when tagged**

- [ ] Atlantoaxial
- [ ] Atlanto-occipital
- [ ] Glenohumeral
- [ ] Acromioclavicular
- [ ] Sternoclavicular
- [ ] Manubriosternal
- [ ] Radioulnar
- [ ] Talocrural
- [ ] Tibiofibular
- [ ] Temporomandibular
- [ ] Sacroiliac
- [ ] Pubic symphysis

- [ ] **A typical synovial joint**

- [ ] Articular cartilage
- [ ] Synovial membrane
- [ ] Joint cavity
- [ ] Synovial fluid

- [ ] **The knee joint**

- [ ] Femur
- [ ] Tibia
- [ ] Fibula
- [ ] Patella
- [ ] Patellar ligament
- [ ] Quadriceps tendon
- [ ] Medial (tibial) collateral ligament
- [ ] Lateral (fibular) collateral ligament
- [ ] Medial meniscus
- [ ] Lateral meniscus
- [ ] Anterior cruciate ligament
- [ ] Posterior cruciate ligament

MODULE 3

Cardiovascular, muscle and blood: heart, vessels, muscle microanatomy

TBL 4 Heart. TBL 5 Blood.

Weeks **7 to 10** Exam **3**, **Wed Oct 21** Lab practical and lecture exam **same day**

Week 7 Lower limb, joints and Exam 2

MODULES 2 AND 3

BEFORE MONDAY, SEP 28

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Lower Extremity
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

MON SEP 28 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Kahoot Review: Bone
- [ ] Lab: Lower Extremity
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, SEP 30

- [ ] Run a gap check on both banks and work the three day plan it builds
  Course tools → Mastery OS
- [ ] Produce five brain dump prompts from this module on blank paper
  Course tools → Brain dump practice
- [ ] Walk the lab structure list for this module and tick what you can name blind
  Course tools → Lab sprints
- [ ] Sit a practice exam before exam day
  Course tools → Practice Exam Builder

WED SEP 30 · EXAM DAY

- [ ] Lab practical
- [ ] Lecture exam

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops

- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 7

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Heart wall layers, pericardium, and internal features Name the three heart wall layers and the pericardial membranes, and locate internal features such as auricles, pectinate muscles, fossa ovalis, and trabeculae carneae.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Heart chambers and septa Identify the four heart chambers and the interatrial and interventricular septa, and state what blood each chamber receives and where it sends it.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Heart valves and what each separates Identify the four heart valves by type and location, and state which two chambers or vessels each one separates and where it prevents backflow.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Chordae tendineae and papillary muscles Identify the chordae tendineae and papillary muscles and describe how they anchor and hold the atrioventricular valves shut.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Cardiac muscle tissue and the intercalated disc Describe cardiac muscle tissue and identify the cardiomyocyte, striations, and intercalated disc with its desmosomes and gap junctions.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Pathway of blood through the heart Trace one drop of blood through the four chambers and four valves of the heart in correct order.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Great vessels and aortic arch Identify the great vessels attached to the base of the heart and state which circuit each one serves.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Coronary circulation Trace coronary circulation from the aorta through the coronary arteries, myocardial capillaries, cardiac veins, and coronary sinus to the right atrium, and identify the major coronary vessels.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Conduction system components and pathway Locate each component of the conduction system and trace the pathway in order from the SA node to the Purkinje fibers.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Nerve supply to the heart Describe the autonomic nerve supply reaching the heart through the cardiac plexus and name the sympathetic and vagal parasympathetic sources.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 7

Recognition is not retrieval

Sit with a labelled diagram of the epidermis and it all looks obvious. Stratum basale at the bottom, corneum at the top, the whole thing makes sense. Close the page and draw it from nothing and something different happens.

The gap between those two experiences is the gap between recognition and retrieval, and it is the single most common reason students are surprised by an exam score. Studying by looking trains recognition. Exams and practicals and patients all ask for retrieval.

You can feel the difference easily. Take any topic you think you know and produce it on blank paper with nothing open. Not a list of terms, the whole structure with relationships and labels. If it comes out complete, you have it. If it comes out as scattered fragments, you had recognition and mistook it for knowledge.

This is the entire logic behind the brain dumps, the drawing tasks and the recall cards. They are not extra work stacked on top of studying. They are the studying. The reading and the videos are how the material gets in. Production is how you find out whether it stayed.

Week 8 The heart, conduction and muscle microanatomy

MODULE 3

BEFORE MONDAY, OCT 5

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Thoracic Anatomy (Heart); Blood Vessel Histology
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

MON OCT 5 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Exam 2 rebuttals + Heart Lecture and Cardiac Conduction
- [ ] Lab: Thoracic Anatomy (Heart); Blood Vessel Histology
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, OCT 7

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Muscle Microanatomy and Sarcomere
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards

- [ ] Bring this workbook

WED OCT 7 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Muscle Microanatomy and Sarcomere
- [ ] Lab: Muscle Microanatomy and Sarcomere
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 8

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Muscle tissue identification Recognize the three muscle tissue types, skeletal, cardiac, and smooth, by striations, cell shape, and number of nuclei.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Three tunics of a vessel wall Name the three tunics of a vessel wall, state the tissue each contains, and explain which layer forms a capillary wall.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
The five vessel types Compare arteries, arterioles, capillaries, venules, and veins by wall structure, direction of flow, and role.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Kinds of artery, capillary, and vein Distinguish elastic and muscular arteries, continuous, fenestrated, and sinusoid capillaries, and identify venous valves and their supporting features.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Circulatory routes, portal systems, and anastomoses Describe the pulmonary and systemic circuits and identify portal systems and anastomoses as alternate circulatory arrangements.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Vessel wall disorders, arterial and venous Describe atherosclerosis and how it changes an artery wall, and name common arterial and venous disorders by the structure each affects.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Fetal circulation, shunts, and adult remnants Identify the fetal vessels and three shunts, state how each reroutes blood, and name the adult remnant each becomes.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Connective tissue coverings Name the epimysium, perimysium, and endomysium and trace how the three sheaths merge into the tendon that anchors muscle to bone.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________
Levels of organization Order the nested levels of skeletal muscle from whole muscle through fascicle, fiber, and myofibril down to the myofilament.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ________

Muscle fiber internal structure

Identify the sarcolemma, sarcoplasm, myonuclei, myofibrils, sarcoplasmic reticulum, terminal cisternae, T tubules, and triad of a muscle fiber.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Sarcomere bands and filaments

Diagram a sarcomere and name the Z disc, A band, I band, H zone, M line, and zone of overlap, stating which filaments occupy each region.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Myofilament and structural proteins

Distinguish thick from thin filaments by protein and anchor point and identify the roles of titin, nebulin, alpha-actinin, myomesin, and dystrophin.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Three muscle tissue types

Compare skeletal, cardiac, and smooth muscle by location, striations, control, and cell features including intercalated discs.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Fascicle arrangement patterns

Identify parallel, fusiform, circular, convergent, and pennate fascicle patterns and give an example muscle for each, relating architecture to power versus range of motion.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Muscle roles and naming

Classify agonist, antagonist, synergist, and fixator roles in a movement and decode a muscle name from its direction, size, shape, location, action, origins, or attachments.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Lever systems

Identify the fulcrum, effort, and load of a lever and classify first-, second-, and third-class levers with a body example of each.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Muscles of facial expression

Identify the muscles of facial expression including occipitofrontalis, orbicularis oculi, orbicularis oris, buccinator, zygomaticus, and platysma on a specimen or model.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Muscles of mastication

Identify the muscles of mastication masseter, temporalis, and the medial and lateral pterygoids on a specimen or model.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Muscles of the neck

Identify the sternocleidomastoid, the scalenes, and the suprahyoid and infrahyoid muscles of the neck on a specimen or model.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Extraocular muscles

Identify the four rectus and two oblique extraocular muscles and the levator palpebrae superioris on a specimen or model.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 8

Stop rereading, start producing

If your study session consists of reading your notes, then reading them again, then reading the parts you are shaky on a third time, you are spending real hours for very little return.

Swap the direction. Instead of putting information in front of your eyes, pull it out of your head and onto paper. Read a section once, close it, and write everything you can. Then open it and fill the gaps in a second colour. That single change turns passive minutes into practice.

It will take longer to get through the same material, and it will feel less pleasant, because producing is harder than recognizing. It is also the reason it works.

A practical version for this course: pick one competency, put your notes away, and produce it on a whiteboard. Draw what can be drawn. Say it out loud as if you were teaching it. Then open the notes and mark what was missing. Ten competencies handled that way will do more for you than a full evening of rereading the whole module.

The point is not that reading is useless. It is that reading is the input step, and most students never get past it.

Week 9 Heart anatomy, muscles and blood

MODULE 3

BEFORE MONDAY, OCT 12

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Muscles of the Face, Chest and Back
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON OCT 12 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Muscles of the Face, Chest and Back
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, OCT 14

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Upper Arm and Anterior Forearm (NAV)

Course tools → Lab sprints

- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

WED OCT 14 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Blood Cell Interactive Lecture
- [ ] Lab: Upper Arm and Anterior Forearm (NAV)
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 9

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Blood composition and plasma
Describe blood as a fluid connective tissue and identify plasma, formed elements, hematocrit, buffy coat, and the plasma proteins albumin, globulins, and fibrinogen.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Formed elements and leukocytes
Identify erythrocytes, platelets, and the five leukocytes on a smear and describe erythrocyte structure and the granulocyte and agranulocyte groups.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Hematopoiesis
Explain where blood cells form and identify the hemocytoblast, megakaryocyte, and the myeloid and lymphoid lines of hematopoiesis.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Blood disorders
Name anemia, sickle cell disease, polycythemia, leukemia, leukopenia, thrombocytopenia, and hemophilia and identify the component each one affects.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Chest and anterior arm muscles
Identify on the cadaver or model the pectoralis major, pectoralis minor, serratus anterior, deltoid, biceps brachii, brachialis, and coracobrachialis with their actions.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Forearm compartments
Identify the anterior flexor and posterior extensor forearm muscles including flexor carpi radialis, palmaris longus, pronator teres, and extensor digitorum and state each compartment action.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Posterior shoulder and rotator cuff
Identify the trapezius, latissimus dorsi, rhomboids, levator scapulae, and the four rotator cuff muscles supraspinatus, infraspinatus, teres minor, and subscapularis with their actions.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Upper-extremity arteries and veins

Trace and identify the subclavian, axillary, brachial, radial, and ulnar arteries and the cephalic, basilic, and median cubital veins on the upper limb.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Upper-extremity nerves

Identify the major nerves of the upper limb including the musculocutaneous, median, ulnar, radial, and axillary nerves and the brachial plexus they arise from.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 9

The exam should not be the first time you test yourself

Think about how strange the usual approach is. Students study for two weeks by reading, then walk into the exam and attempt retrieval under time pressure for the first time. The exam becomes both the test and the first practice.

You would not prepare for anything else that way. Nobody learns to place an IV by reading about it and then doing it once, for real, for a grade.

Everything in this course is set up so that the exam is not your first attempt. The brain dumps are retrieval under a clock. The recall cards are retrieval in small doses. The practice exam builder lets you sit a version of the real thing before the real thing. The lab sprints make you name structures without the labels.

Three days before each exam, run a gap check on both banks and let it build you a plan. Then work the plan rather than rereading the module from the top. The goal is that by exam day you have already answered questions like these, found out what you did not have, and fixed it.

An exam should confirm what you already know about yourself, not tell you something new.

Week 10 Blood, forearm structures and Exam 3

MODULE 3

BEFORE MONDAY, OCT 19

- ☐ Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- ☐ Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- ☐ Pre-study the lab structures for Posterior Forearm (NAV)
  Course tools → Lab sprints
- ☐ Run the recall cards that are due

Course tools → Recall cards

- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON OCT 19 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Posterior Forearm (NAV)
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, OCT 21

- [ ] Run a gap check on both banks and work the three day plan it builds
  Course tools → Mastery OS
- [ ] Produce five brain dump prompts from this module on blank paper
  Course tools → Brain dump practice
- [ ] Walk the lab structure list for this module and tick what you can name blind
  Course tools → Lab sprints
- [ ] Sit a practice exam before exam day
  Course tools → Practice Exam Builder

WED OCT 21 · EXAM DAY

- [ ] Lab practical
- [ ] Lecture exam

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS
- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 10

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

No new competencies open this week. Use the loop on anything from the module you are still shaky on.

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

Something that still needs work

My priority for spaced retrieval next week

LEARNING TO LEARN · WEEK 10

What experts do when they do not know

One thing that separates an experienced clinician from a student is not that the clinician knows everything. It is what they do at the edge of what they know.

A student who hits an unfamiliar structure usually stalls, guesses, or looks for someone to hand them the answer. An expert reasons from what they do know. What is around it? What does it look like? What kind of tissue is that? What would something in this location plausibly connect to? They narrow the possibilities before they look anything up, and when they do look it up, they have a specific question rather than a vague one.

You can practice this deliberately. When you meet a structure you cannot name on a slide or in the lab, do not go straight to the key. Say what you can see. Say what it rules out. Commit to a best guess and state why. Then check.

This is also what I am asking for when I say to bring your attempts when you ask for help. Not because the attempt has to be right, but because reasoning to the edge of your knowledge is the skill, and it only develops if you spend time there.

Lab structures Module 3

EXAM 3 PRACTICAL

Tick the block name once you can identify it on the slide or the model, fill in function and location where the lines are there, then tick each structure you can point to. Anything not on this list is context, not something you will be asked to identify. The reference tables, regional and directional terms, quadrants and the rest, are in the standalone Module 3 structure list on the course site.

Muscle Tissue and Microanatomy

- [ ] **Skeletal muscle, longitudinal section**

FUNCTION ______________________ LOCATION ______________________

- [ ] Muscle fiber
- [ ] Sarcolemma
- [ ] Sarcoplasm
- [ ] Peripheral nuclei
- [ ] Striations
- [ ] Myofibrils
- [ ] Endomysium

- [ ] **Skeletal muscle, cross section**

- [ ] Muscle fiber, cut across
- [ ] Fascicle
- [ ] Endomysium
- [ ] Perimysium
- [ ] Epimysium
- [ ] Peripheral nuclei

- [ ] **Cardiac muscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Cardiomyocyte
- [ ] Branching fibers
- [ ] Central nucleus
- [ ] Intercalated discs
- [ ] Striations

- [ ] **Smooth muscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Spindle-shaped cell
- [ ] Central nucleus
- [ ] Absence of striations
- [ ] Longitudinal layer
- [ ] Circular layer

- [ ] **The muscle organ model**

- [ ] Epimysium
- [ ] Perimysium
- [ ] Endomysium
- [ ] Fascicle
- [ ] Muscle fiber
- [ ] Tendon
- [ ] Aponeurosis
- [ ] Deep fascia
- [ ] Belly
- [ ] Origin
- [ ] Insertion

- [ ] **The muscle fiber model**

- [ ] Sarcolemma - [ ] Sarcoplasm - [ ] Myonuclei - [ ] Myofibril - [ ] Transverse (T) tubule - [ ] Sarcoplasmic reticulum - [ ] Terminal cisternae - [ ] Triad - [ ] Mitochondria

- [ ] **The sarcomere model**

- [ ] Z disc - [ ] A band - [ ] I band - [ ] H zone - [ ] M line - [ ] Zone of overlap - [ ] Thick filament (myosin) - [ ] Thin filament (actin) - [ ] Titin - [ ] Cross bridges

- [ ] **Fascicle arrangement, on models**

- [ ] Parallel - [ ] Fusiform - [ ] Circular - [ ] Convergent - [ ] Unipennate - [ ] Bipennate - [ ] Multipennate

- [ ] **Lever systems**

- [ ] Fulcrum - [ ] Effort - [ ] Load - [ ] First class lever - [ ] Second class lever - [ ] Third class lever

Muscles of the Back and Thorax

- [ ] **Superficial back**

- [ ] Trapezius - [ ] Latissimus dorsi - [ ] Levator scapulae - [ ] Rhomboid major - [ ] Rhomboid minor - [ ] Thoracolumbar fascia - [ ] Ligamentum nuchae

- [ ] **Erector spinae**

- [ ] Iliocostalis - [ ] Longissimus - [ ] Spinalis

- [ ] **Deep back**

- [ ] Splenius capitis - [ ] Splenius cervicis - [ ] Semispinalis - [ ] Multifidus - [ ] Quadratus lumborum - [ ] Rotatores - [ ] Interspinales - [ ] Intertransversarii

- [ ] **Thoracic wall**

- [ ] External intercostals - [ ] Internal intercostals - [ ] Innermost intercostals - [ ] Transversus thoracis - [ ] Serratus posterior superior - [ ] Serratus posterior inferior - [ ] Intercostal space - [ ] Neurovascular bundle in the costal groove

- [ ] **The diaphragm**

- [ ] Central tendon - [ ] Right crus - [ ] Left crus - [ ] Costal part - [ ] Sternal part - [ ] Caval opening (T8) - [ ] Esophageal hiatus (T10) - [ ] Aortic hiatus (T12) - [ ] Phrenic nerve

- [ ] **Anterior thorax**

- [ ] Pectoralis major - [ ] Pectoralis minor - [ ] Serratus anterior - [ ] Subclavius - [ ] External oblique, upper slips - [ ] Clavipectoral fascia

Muscles of the Upper Extremity

- [ ] **Rotator cuff**

- [ ] Supraspinatus - [ ] Infraspinatus - [ ] Teres minor - [ ] Subscapularis

- [ ] **Scapulohumeral, other**

- [ ] Deltoid, anterior fibers - [ ] Deltoid, middle fibers - [ ] Deltoid, posterior fibers - [ ] Teres major - [ ] Coracobrachialis

- [ ] **Arm, anterior compartment**

- [ ] Biceps brachii, long head - [ ] Biceps brachii, short head - [ ] Bicipital aponeurosis - [ ] Brachialis - [ ] Coracobrachialis

- [ ] **Arm, posterior compartment**

- [ ] Triceps brachii, long head - [ ] Triceps brachii, lateral head - [ ] Triceps brachii, medial head - [ ] Anconeus - [ ] Radial groove

☐ Forearm, anterior compartment
☐ Brachioradialis ☐ Pronator teres ☐ Flexor carpi radialis ☐ Palmaris longus ☐ Flexor carpi ulnaris
☐ Flexor digitorum superficialis ☐ Flexor digitorum profundus ☐ Flexor pollicis longus ☐ Pronator quadratus
☐ Medial epicondyle, common flexor origin

☐ Forearm, posterior compartment
☐ Extensor carpi radialis longus ☐ Extensor carpi radialis brevis ☐ Extensor digitorum ☐ Extensor digiti minimi
☐ Extensor carpi ulnaris ☐ Supinator ☐ Abductor pollicis longus ☐ Extensor pollicis brevis ☐ Extensor pollicis longus
☐ Extensor indicis ☐ Lateral epicondyle, common extensor origin

☐ Hand, intrinsic muscles
☐ Thenar eminence ☐ Abductor pollicis brevis ☐ Flexor pollicis brevis ☐ Opponens pollicis ☐ Adductor pollicis
☐ Hypothenar eminence ☐ Abductor digiti minimi ☐ Flexor digiti minimi brevis ☐ Opponens digiti minimi ☐ Lumbricals
☐ Palmar interossei ☐ Dorsal interossei

☐ Landmarks and retinacula
☐ Cubital fossa ☐ Carpal tunnel ☐ Flexor retinaculum ☐ Extensor retinaculum ☐ Palmar aponeurosis
☐ Anatomical snuffbox ☐ Extensor expansion

The Heart

☐ Position and coverings
☐ Mediastinum ☐ Apex ☐ Base ☐ Anterior (sternocostal) surface ☐ Inferior (diaphragmatic) surface
☐ Fibrous pericardium ☐ Parietal serous pericardium ☐ Visceral serous pericardium (epicardium) ☐ Pericardial cavity

☐ The heart wall
FUNCTION ____________________ LOCATION ____________________

☐ Epicardium ☐ Myocardium ☐ Endocardium

☐ External surface features
☐ Coronary sulcus ☐ Anterior interventricular sulcus ☐ Posterior interventricular sulcus ☐ Right auricle ☐ Left auricle
☐ Ligamentum arteriosum

☐ Chambers and internal features
☐ Right atrium ☐ Right ventricle ☐ Left atrium ☐ Left ventricle ☐ Interatrial septum ☐ Interventricular septum
☐ Fossa ovalis ☐ Pectinate muscles ☐ Crista terminalis ☐ Sinus venarum ☐ Trabeculae carneae ☐ Papillary muscles
☐ Chordae tendineae ☐ Moderator band ☐ Conus arteriosus ☐ Aortic vestibule

☐ Valves
☐ Tricuspid valve ☐ Mitral (bicuspid) valve ☐ Pulmonary semilunar valve ☐ Aortic semilunar valve ☐ Fibrous skeleton

☐ Great vessels
☐ Superior vena cava ☐ Inferior vena cava ☐ Pulmonary trunk ☐ Right pulmonary artery ☐ Left pulmonary artery
☐ Pulmonary veins ☐ Ascending aorta ☐ Aortic arch ☐ Descending thoracic aorta

☐ Coronary circulation
☐ Right coronary artery ☐ Right marginal artery ☐ Posterior interventricular artery (PDA) ☐ Left coronary artery
☐ Anterior interventricular artery (LAD) ☐ Circumflex artery ☐ Left marginal artery ☐ Coronary sinus ☐ Great cardiac vein
☐ Middle cardiac vein ☐ Small cardiac vein

☐ Conduction system, on the model
☐ SA node ☐ AV node ☐ AV bundle (bundle of His) ☐ Right bundle branch ☐ Left bundle branch ☐ Purkinje fibers

- [ ] **Cardiac muscle slide**

FUNCTION ______________________ LOCATION ______________________

- [ ] Cardiomyocyte - [ ] Central nucleus - [ ] Branching fibers - [ ] Intercalated discs - [ ] Striations

Blood Vessels, Histology

- [ ] **Artery, cross section**

FUNCTION ______________________ LOCATION ______________________

- [ ] Tunica intima - [ ] Endothelium - [ ] Basement membrane - [ ] Internal elastic lamina - [ ] Tunica media - [ ] Smooth muscle
- [ ] External elastic lamina - [ ] Tunica externa - [ ] Vasa vasorum - [ ] Lumen

- [ ] **Vein, cross section**

FUNCTION ______________________ LOCATION ______________________

- [ ] Tunica intima - [ ] Tunica media - [ ] Tunica externa - [ ] Venous valve - [ ] Lumen

- [ ] **Elastic artery**
- [ ] Elastic lamellae - [ ] Thick tunica media - [ ] Internal elastic lamina - [ ] External elastic lamina

- [ ] **Muscular artery**
- [ ] Prominent internal elastic lamina - [ ] Smooth muscle of the tunica media - [ ] Thin external elastic lamina

- [ ] **Arteriole and capillary bed**
- [ ] Arteriole - [ ] Metarteriole - [ ] Precapillary sphincter - [ ] Capillary - [ ] Capillary bed - [ ] Thoroughfare channel
- [ ] Postcapillary venule - [ ] Muscular venule

- [ ] **Capillary types**
- [ ] Continuous capillary - [ ] Fenestrated capillary - [ ] Sinusoid - [ ] Intercellular cleft - [ ] Fenestration
- [ ] Basement membrane

- [ ] **Fetal circulation, on the model**
- [ ] Placenta - [ ] Umbilical vein - [ ] Ductus venosus - [ ] Foramen ovale - [ ] Ductus arteriosus - [ ] Umbilical arteries

Arteries and Veins of the Thorax and Upper Limb

- [ ] **Aortic arch and its branches**
- [ ] Ascending aorta - [ ] Aortic arch - [ ] Brachiocephalic trunk - [ ] Right common carotid artery - [ ] Right subclavian artery
- [ ] Left common carotid artery - [ ] Left subclavian artery - [ ] Descending thoracic aorta

- [ ] **Other thoracic arteries**
- [ ] Internal thoracic artery - [ ] Anterior intercostal arteries - [ ] Posterior intercostal arteries - [ ] Bronchial arteries
- [ ] Esophageal arteries - [ ] Vertebral artery - [ ] Thyrocervical trunk - [ ] Costocervical trunk

- [ ] **Arteries of the upper limb**
- [ ] Subclavian artery - [ ] Axillary artery - [ ] Thoracoacromial artery - [ ] Lateral thoracic artery - [ ] Subscapular artery
- [ ] Anterior circumflex humeral artery - [ ] Posterior circumflex humeral artery - [ ] Brachial artery - [ ] Deep brachial artery
- [ ] Radial artery - [ ] Ulnar artery - [ ] Common interosseous artery - [ ] Superficial palmar arch - [ ] Deep palmar arch
- [ ] Digital arteries

- [ ] **Superficial veins of the upper limb**
- [ ] Cephalic vein - [ ] Basilic vein - [ ] Median cubital vein - [ ] Median antebrachial vein - [ ] Dorsal venous network of the hand

- [ ] **Deep veins of the thorax and upper limb**

- [ ] Radial veins - [ ] Ulnar veins - [ ] Brachial veins - [ ] Axillary vein - [ ] Subclavian vein - [ ] Internal jugular vein
- [ ] Brachiocephalic veins - [ ] Superior vena cava - [ ] Azygos vein - [ ] Hemiazygos vein - [ ] Internal thoracic vein

Nerves of the Upper Limb

- [ ] **Levels of the plexus**

- [ ] Roots, C5 to T1 - [ ] Upper trunk - [ ] Middle trunk - [ ] Lower trunk - [ ] Anterior divisions - [ ] Posterior divisions
- [ ] Lateral cord - [ ] Posterior cord - [ ] Medial cord

- [ ] **Terminal branches**

- [ ] Musculocutaneous nerve - [ ] Median nerve - [ ] Ulnar nerve - [ ] Radial nerve - [ ] Axillary nerve

- [ ] **Branches above the terminal nerves**

- [ ] Dorsal scapular nerve - [ ] Long thoracic nerve - [ ] Suprascapular nerve - [ ] Nerve to subclavius - [ ] Lateral pectoral nerve
- [ ] Medial pectoral nerve - [ ] Thoracodorsal nerve - [ ] Upper subscapular nerve - [ ] Lower subscapular nerve
- [ ] Medial cutaneous nerve of the arm - [ ] Medial cutaneous nerve of the forearm

- [ ] **Nerves at their vulnerable points**

- [ ] Axillary nerve at the surgical neck of the humerus - [ ] Radial nerve in the radial groove
- [ ] Ulnar nerve behind the medial epicondyle - [ ] Median nerve in the carpal tunnel - [ ] Long thoracic nerve on the chest wall

- [ ] **Thoracic nerves**

- [ ] Intercostal nerves - [ ] Phrenic nerve - [ ] Vagus nerve in the thorax - [ ] Sympathetic trunk

Blood

- [ ] **Whole blood, in a spun tube**

- [ ] Plasma layer - [ ] Buffy coat - [ ] Erythrocyte layer - [ ] Hematocrit

- [ ] **Erythrocytes**

FUNCTION ______________________ LOCATION ______________________

- [ ] Biconcave disc - [ ] Pale central area - [ ] Absence of a nucleus - [ ] Rouleaux formation

- [ ] **Granulocytes**

- [ ] Neutrophil - [ ] Multi-lobed nucleus - [ ] Fine pale granules - [ ] Eosinophil - [ ] Bilobed nucleus
- [ ] Coarse red-orange granules - [ ] Basophil - [ ] Nucleus obscured by dark granules

- [ ] **Agranulocytes**

- [ ] Lymphocyte - [ ] Large round nucleus with thin rim of cytoplasm - [ ] Monocyte - [ ] Kidney-shaped nucleus
- [ ] Abundant pale cytoplasm

- [ ] **Platelets and formation**

- [ ] Platelet (thrombocyte) - [ ] Megakaryocyte - [ ] Red bone marrow - [ ] Hemocytoblast - [ ] Reticulocyte

MODULE 4

Respiratory, lymphatic and immune, digestive, endocrine

34 competencies. TBL 6 Respiratory. TBL 7 GI.

Weeks **11 to 14** Exam **4, Mon Nov 16** Lab practical and lecture exam **same day**

Week 11 Respiratory, endocrine and thigh structures

BEFORE MONDAY, OCT 26

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Respiratory Anatomy and Histology
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

MON OCT 26 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Respiratory Anatomy and Histology Walk-Through
- [ ] Lab: Respiratory Anatomy and Histology
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, OCT 28

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course tools → Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course tools → Course materials
- [ ] Pre-study the lab structures for Anterior Thigh Muscles and NAV
  Course tools → Lab sprints
- [ ] Run the recall cards that are due
  Course tools → Recall cards
- [ ] Bring this workbook

WED OCT 28 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Endocrine Walk-Through
- [ ] Lab: Anterior Thigh Muscles and NAV
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Course tools → Mastery OS

- [ ] Loops for the lab material
  Course tools → Loops
- [ ] One prompt from an earlier module, produced blind
  Course tools → Brain dump practice

Competency loop, week 11

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Upper and lower tracts, conducting and respiratory zones Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Nose, nasal cavity, and pharynx Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Larynx Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Trachea and bronchial tree Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Lungs and pleurae Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Respiratory zone and alveolar structure Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Thoracic wall and diaphragm Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Respiratory disorders by structure affected Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Lymph pathway Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Vessel and node structure Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Primary and secondary organs Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Lymphatic disorders Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______
Abdominal and pelvic wall muscles Identify the abdominal and pelvic wall muscles on the cadaver including the rectus abdominis, external and internal obliques, transversus abdominis, and the pelvic floor muscles.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Gluteal and thigh muscles

Identify the gluteal and thigh muscles including the gluteal group, quadriceps, hamstrings, and adductor group on the cadaver.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Leg muscles

Identify the muscles of the leg including the anterior, lateral, and posterior compartment muscles on the cadaver.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Major endocrine glands and locations

Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Pituitary gross anatomy and lobes

Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Thyroid and parathyroid anatomy

Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Adrenal gland zones

Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

Pancreas, gonads, and organs with endocrine tissue

Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 11

More hours does not always mean better studying

Two students spend six hours on Module 3. One finishes able to reconstruct the conduction pathway from nothing. The other finishes with beautifully highlighted notes and no ability to produce anything without them. The hours were the same.

Time is not the input that matters. What you did with the time is.

If you are putting in real hours and your exam scores are not moving, adding more hours is usually the wrong fix. Change what the hours contain. Look honestly at a study session and ask how many minutes of it involved producing something from memory. For most students the answer is close to zero, and that is the whole problem.

There is also a limit worth respecting. Attention degrades, and the last hour of a five hour session is rarely worth much. Shorter sessions spread across more days beat one long session, for the same total time, because the spacing itself does work for you.

Try this: cut your session length, raise the number of sessions, and make sure every session includes at least one thing you produced with everything closed.

Week 12 Respiratory TBL and the GI map

MODULE 4

BEFORE MONDAY, NOV 2

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for Posterior Thigh Muscle and NAV
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON NOV 2 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: Posterior Thigh Muscle and NAV
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, NOV 4

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials

- [ ] Pre-study the lab structures for Anterior/Posterior Lower-Leg Muscles
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Bring this workbook

WED NOV 4 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] GI Map Activity I and II
- [ ] Lab: Anterior/Posterior Lower-Leg Muscles
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind
  Brain dump practice

Competency loop, week 12

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Four-layer gut wall Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Enteric nerve plexuses Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Peritoneum and folds Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Retroperitoneal organs Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Mouth, pharynx, esophagus Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Major canal sphincters Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Stomach regions and features Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Small intestine and surface area Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Large intestine features
Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Trace the pathway of food
Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Clinical anatomy of canal disorders
Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 12

Build connections, not piles of facts

By this point in the term you have a lot of individual facts. The risk is that they stay individual, sitting in separate piles labelled tissues, bones, heart, muscles.

Anatomy does not work that way and neither do exams past the first one. A question about the respiratory system is also a question about epithelium, because the tissue lining changes as you move down the tract and the change is the point. A question about a joint is a question about cartilage and connective tissue proper. The systems keep reusing the same building blocks.

Facts that connect to other facts are easier to retrieve, because you have more than one route in. Facts sitting alone have exactly one route, and if you miss it you have nothing.

So spend some of your study time deliberately building links. Take something from this week and something from Module 1 and write how they relate. Draw a map with a system in the middle and every tissue type it uses hanging off it. When you learn a new structure, ask what it is made of and what it resembles that you have already seen.

The map is worth more than the pile.

Week 13 The GI system

MODULE 4

BEFORE MONDAY, NOV 9

☐ Work the pre-work sheet with your notes packet and the reading open beside you
Course materials

- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for GI Organs (Primary and Accessory)
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON NOV 9 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: GI Organs (Primary and Accessory)
- [ ] Round tables, practice rounds

WED NOV 11 · CAMPUS CLOSED

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind
  Brain dump practice

Competency loop, week 13

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Teeth and tongue Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Salivary glands Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Liver gross and lobule anatomy Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Gallbladder and bile pathway Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Pancreas structure Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______
Clinical anatomy of accessory disorders Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.	☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Abdominal aorta and portal system

Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Lower limb vessels and nerves

Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 13

Why drawing shows you what words hide

You can write a fluent paragraph about the layers of an artery and still not know how they sit relative to each other. Words let you be vague. A drawing does not.

When you draw a structure from memory you have to commit to everything at once. What is inside what. What touches what. Which side the thing is on. How many of them there are. Every one of those is a decision you can dodge in a sentence and cannot dodge in a picture.

That is why the drawing tasks in this course come with a checklist rather than a finished image to copy. Copying a diagram is recognition again. Producing one and then scoring yourself against a list of what should be there is retrieval, and the gap between what you drew and what the list wanted is the most specific study information you will get all week.

Your drawings do not need to be good. Nobody is grading the art. A rough sketch with correct relationships is worth more than a beautiful one with the layers in the wrong order.

If a topic feels solid to you, draw it blind before you believe that.

Week 14 Exam 4, then renal anatomy

MODULES 4 AND 5

Week 14 sits in two modules. Exam 4 closes Module 4 this week, and the teaching that starts Module 5 begins in the same week. The full week page, with the checklists and the competency loop, is printed at the front of Module 5. Work it there.

Lab structures Module 4

EXAM 4 PRACTICAL

Tick the block name once you can identify it on the slide or the model, fill in function and location where the lines are there, then tick each structure you can point to. Anything not on this list is context, not something you will be asked to identify. The reference tables, regional and directional terms, quadrants and the rest, are in the standalone Module 4 structure list on the course site.

The Respiratory System

☐ External nose and nasal cavity

☐ Root ☐ Bridge ☐ Dorsum nasi ☐ Apex ☐ Nares (nostrils) ☐ Nasal vestibule ☐ Nasal cavity ☐ Nasal septum
☐ Septal cartilage ☐ Perpendicular plate of the ethmoid ☐ Vomer ☐ Superior nasal concha ☐ Middle nasal concha
☐ Inferior nasal concha ☐ Superior meatus ☐ Middle meatus ☐ Inferior meatus ☐ Hard palate ☐ Soft palate
☐ Choanae

☐ Paranasal sinuses

☐ Frontal sinus ☐ Ethmoidal air cells ☐ Sphenoidal sinus ☐ Maxillary sinus

☐ Pharynx

☐ Nasopharynx ☐ Pharyngeal tonsil ☐ Opening of the auditory tube ☐ Oropharynx ☐ Palatine tonsil ☐ Lingual tonsil
☐ Laryngopharynx ☐ Uvula ☐ Soft palate

☐ Larynx

☐ Thyroid cartilage ☐ Laryngeal prominence ☐ Cricoid cartilage ☐ Epiglottis ☐ Arytenoid cartilages
☐ Corniculate cartilages ☐ Cuneiform cartilages ☐ Cricothyroid ligament ☐ Vocal folds (true vocal cords)
☐ Vocal ligaments ☐ Vestibular folds (false vocal cords) ☐ Glottis ☐ Hyoid bone

☐ Trachea and bronchial tree

☐ Trachea ☐ Tracheal cartilages (C rings) ☐ Trachealis muscle ☐ Carina ☐ Right main (primary) bronchus
☐ Left main (primary) bronchus ☐ Lobar (secondary) bronchi ☐ Segmental (tertiary) bronchi ☐ Bronchioles
☐ Terminal bronchioles

☐ Lungs and pleurae

☐ Right lung, superior lobe ☐ Right lung, middle lobe ☐ Right lung, inferior lobe ☐ Horizontal fissure ☐ Oblique fissure
☐ Left lung, superior lobe ☐ Left lung, inferior lobe ☐ Cardiac notch ☐ Lingula ☐ Apex of the lung ☐ Base of the lung
☐ Hilum of the right lung ☐ Hilum of the left lung ☐ Root of the lung ☐ Parietal pleura ☐ Visceral pleura
☐ Right pleural cavity ☐ Left pleural cavity

☐ The thorax on the cadaver

☐ Thoracic cavity ☐ Mediastinum ☐ Esophagus ☐ Diaphragm ☐ External intercostals ☐ Internal intercostals

☐ Vessels and nerves of the thorax

☐ Brachiocephalic trunk ☐ Right subclavian artery ☐ Left subclavian artery ☐ Right common carotid artery
☐ Left common carotid artery ☐ Right phrenic nerve ☐ Left phrenic nerve ☐ Right brachiocephalic vein
☐ Left brachiocephalic vein ☐ Right subclavian vein ☐ Left subclavian vein ☐ Right internal jugular vein
☐ Left internal jugular vein ☐ Right pulmonary artery ☐ Left pulmonary artery ☐ Right pulmonary veins
☐ Left pulmonary veins

☐ The respiratory zone and the alveolus

FUNCTION ______________________ **LOCATION** ______________________

☐ Respiratory bronchiole ☐ Alveolar duct ☐ Alveolar sac ☐ Alveolus ☐ Type I alveolar cell ☐ Type II alveolar cell
☐ Alveolar macrophage ☐ Respiratory membrane ☐ Pulmonary capillary

- [ ] **Respiratory slides**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Pseudostratified ciliated columnar epithelium
- [ ] Goblet cells
- [ ] Cilia
- [ ] Hyaline cartilage of the trachea
- [ ] Simple squamous epithelium of the alveolus
- [ ] Lung tissue, low power

The Lymphatic System

- [ ] **Lymphatic vessels, from the tissue spaces to the veins**

- [ ] Lymphatic capillaries
- [ ] Lacteals
- [ ] Lymphatic collecting vessels
- [ ] Lymphatic valves
- [ ] Lymphatic trunks
- [ ] Right lymphatic duct
- [ ] Thoracic duct
- [ ] Cisterna chyli
- [ ] Subclavian veins
- [ ] Internal jugular veins

- [ ] **The lymph node**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Capsule
- [ ] Cortex
- [ ] Medulla
- [ ] Afferent lymphatic vessels
- [ ] Efferent lymphatic vessels
- [ ] Hilum
- [ ] Cervical nodes
- [ ] Axillary nodes
- [ ] Inguinal nodes

- [ ] **The lymphatic organs**

- [ ] Red bone marrow
- [ ] Thymus
- [ ] Lymph nodes
- [ ] Spleen
- [ ] Tonsils
- [ ] MALT

- [ ] **The spleen**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Spleen
- [ ] White pulp
- [ ] Red pulp

- [ ] **The tonsils**

- [ ] Pharyngeal tonsil
- [ ] Palatine tonsil
- [ ] Lingual tonsil

The Alimentary Canal

- [ ] **Body cavities, serous membranes, and the omenta**

- [ ] Abdominal cavity
- [ ] Pelvic cavity
- [ ] Peritoneal cavity
- [ ] Parietal peritoneum
- [ ] Visceral peritoneum
- [ ] Retroperitoneum
- [ ] Greater omentum
- [ ] Lesser omentum
- [ ] Falciform ligament
- [ ] Mesentery
- [ ] Mesocolon

- [ ] **The wall of the canal, in section**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Mucosa
- [ ] Epithelium
- [ ] Lamina propria
- [ ] Muscularis mucosae
- [ ] Submucosa
- [ ] Submucosal plexus
- [ ] Muscularis, inner circular layer
- [ ] Muscularis, outer longitudinal layer
- [ ] Myenteric plexus
- [ ] Serosa
- [ ] Adventitia
- [ ] Lumen

- [ ] **Mouth, pharynx, and esophagus**

- [ ] Oral cavity
- [ ] Hard palate
- [ ] Soft palate
- [ ] Uvula
- [ ] Fauces
- [ ] Palatine tonsil
- [ ] Lingual tonsil
- [ ] Nasopharynx
- [ ] Oropharynx
- [ ] Laryngopharynx
- [ ] Esophagus
- [ ] Esophageal hiatus
- [ ] Upper esophageal sphincter
- [ ] Lower (cardiac) esophageal sphincter

- [ ] **The stomach**

- [ ] Cardiac region
- [ ] Fundus
- [ ] Body
- [ ] Pyloric region
- [ ] Pyloric antrum
- [ ] Pyloric canal
- [ ] Pylorus
- [ ] Pyloric sphincter
- [ ] Cardiac sphincter
- [ ] Lesser curvature
- [ ] Greater curvature
- [ ] Gastric rugae
- [ ] Gastric pits
- [ ] Gastric glands
- [ ] Muscularis, oblique layer
- [ ] Muscularis, circular layer
- [ ] Muscularis, longitudinal layer

- [ ] **The small intestine**

FUNCTION ______________________ LOCATION ______________________

- [ ] Duodenum - [ ] Major duodenal papilla - [ ] Jejunum - [ ] Ileum - [ ] Ileocecal valve - [ ] Ileocecal sphincter - [ ] Mesentery
- [ ] Circular folds (plicae circulares) - [ ] Villi - [ ] Lacteal - [ ] Microvilli (brush border)

- [ ] **The large intestine**

- [ ] Cecum - [ ] Vermiform appendix - [ ] Ascending colon - [ ] Hepatic flexure (right colic flexure) - [ ] Transverse colon
- [ ] Splenic flexure (left colic flexure) - [ ] Descending colon - [ ] Sigmoid colon - [ ] Rectum - [ ] Anal canal
- [ ] Internal anal sphincter - [ ] External anal sphincter - [ ] Teniae coli - [ ] Haustra - [ ] Omental appendices

The Accessory Digestive Organs

- [ ] **The tongue and the salivary glands**

- [ ] Tongue - [ ] Lingual frenulum - [ ] Papillae - [ ] Intrinsic tongue muscles - [ ] Extrinsic tongue muscles - [ ] Parotid gland
- [ ] Submandibular gland - [ ] Sublingual gland

- [ ] **The liver**

- [ ] Right lobe - [ ] Left lobe - [ ] Quadrate lobe - [ ] Caudate lobe - [ ] Falciform ligament
- [ ] Ligamentum teres (round ligament) - [ ] Lesser omentum

- [ ] **Liver tissue, on the slide and the model**

FUNCTION ______________________ LOCATION ______________________

- [ ] Hepatocytes - [ ] Hepatic laminae - [ ] Hepatic sinusoids - [ ] Bile canaliculi - [ ] Portal triad - [ ] Bile duct
- [ ] Branch of the hepatic artery - [ ] Branch of the hepatic portal vein - [ ] Central vein

- [ ] **The gallbladder and the biliary tree**

- [ ] Gallbladder - [ ] Fundus of the gallbladder - [ ] Body of the gallbladder - [ ] Neck of the gallbladder - [ ] Right hepatic duct
- [ ] Left hepatic duct - [ ] Common hepatic duct - [ ] Cystic duct - [ ] Common bile duct - [ ] Hepatopancreatic ampulla
- [ ] Major duodenal papilla - [ ] Sphincter of Oddi

- [ ] **The pancreas**

- [ ] Head of the pancreas - [ ] Body of the pancreas - [ ] Tail of the pancreas - [ ] Main pancreatic duct
- [ ] Accessory pancreatic duct - [ ] Hepatopancreatic ampulla - [ ] Major duodenal papilla - [ ] Acini - [ ] Pancreatic islets

The Urinary System

- [ ] **Gross anatomy of the urinary system**

- [ ] Right kidney - [ ] Left kidney - [ ] Suprarenal (adrenal) gland - [ ] Renal capsule - [ ] Renal cortex - [ ] Renal medulla
- [ ] Renal column - [ ] Renal pyramid - [ ] Renal papilla - [ ] Minor calyx - [ ] Major calyx - [ ] Renal pelvis - [ ] Renal sinus
- [ ] Renal hilum - [ ] Ureter - [ ] Urinary bladder - [ ] Urethra

- [ ] **Vasculature of the kidney**

- [ ] Renal artery - [ ] Arcuate artery - [ ] Cortical radiate artery - [ ] Afferent arteriole - [ ] Efferent arteriole
- [ ] Peritubular capillaries - [ ] Vasa recta - [ ] Cortical radiate vein - [ ] Arcuate vein - [ ] Renal vein

- [ ] **The renal corpuscle**

FUNCTION ______________________ LOCATION ______________________

- [ ] Glomerulus - [ ] Glomerular (Bowman's) capsule - [ ] Parietal layer of the glomerular capsule
- [ ] Visceral layer of the glomerular capsule - [ ] Podocytes - [ ] Capsular space

- [ ] **The renal tubule and the juxtaglomerular apparatus**

- [ ] Proximal convoluted tubule - [ ] Loop of Henle, descending limb - [ ] Loop of Henle, ascending limb
- [ ] Distal convoluted tubule - [ ] Collecting duct - [ ] Papillary duct - [ ] Juxtaglomerular apparatus - [ ] Macula densa cells
- [ ] Granular (juxtaglomerular) cells

- [ ] **The bladder and urethra**

- [ ] Trigone - [ ] Ureteral openings - [ ] Internal urethral orifice - [ ] Detrusor muscle - [ ] Internal urethral sphincter
- [ ] External urethral sphincter - [ ] Male urethra - [ ] Female urethra

The Male Reproductive System

- [ ] **The scrotum and the testis**

- [ ] Scrotum - [ ] Testis - [ ] Cremaster muscle - [ ] Dartos muscle - [ ] Tunica vaginalis - [ ] Tunica albuginea
- [ ] Lobules of the testis - [ ] Seminiferous tubules - [ ] Testicular artery - [ ] Testicular vein - [ ] Pampiniform plexus
- [ ] Spermatic cord - [ ] Inguinal canal

- [ ] **Testis tissue, on the slide**

FUNCTION ______________________ LOCATION ______________________

- [ ] Seminiferous tubule - [ ] Sustentacular (Sertoli) cells - [ ] Interstitial (Leydig) cells - [ ] Developing sperm cells

- [ ] **The duct system**

- [ ] Epididymis - [ ] Vas deferens (ductus deferens) - [ ] Ejaculatory duct - [ ] Prostatic urethra - [ ] Membranous urethra
- [ ] Spongy (penile) urethra

- [ ] **The accessory glands**

- [ ] Seminal vesicles - [ ] Prostate gland - [ ] Bulbourethral glands

- [ ] **The penis**

- [ ] Root - [ ] Body - [ ] Glans penis - [ ] Prepuce - [ ] Corpus spongiosum - [ ] Corpora cavernosa - [ ] Spongy (penile) urethra

- [ ] **The pelvic vessels**

- [ ] Internal iliac artery - [ ] External iliac artery - [ ] Medial umbilical ligament - [ ] Internal iliac vein - [ ] External iliac vein

The Female Reproductive System

- [ ] **The ovary and its attachments**

- [ ] Ovary - [ ] Ovarian ligament - [ ] Suspensory ligament - [ ] Broad ligament of the uterus - [ ] Mesovarium - [ ] Ovarian artery
- [ ] Ovarian vein

- [ ] **Ovary tissue, on the slide**

FUNCTION ______________________ LOCATION ______________________

- [ ] Germinal epithelium - [ ] Tunica albuginea - [ ] Ovarian cortex - [ ] Ovarian medulla - [ ] Ovarian follicles - [ ] Mature follicle
- [ ] Corpus luteum - [ ] Corpus albicans

- [ ] **The uterine tubes**

- [ ] Fimbriae - [ ] Infundibulum - [ ] Ampulla - [ ] Isthmus

- [ ] **The uterus**

- [ ] Fundus - [ ] Body - [ ] Cervix - [ ] Uterine cavity - [ ] Cervical canal - [ ] Round ligament of the uterus
- [ ] Broad ligament of the uterus - [ ] Uterosacral ligament - [ ] Cardinal ligament - [ ] Perimetrium - [ ] Myometrium
- [ ] Endometrium

☐ The vagina and the vulva

☐ Vagina ☐ Vaginal orifice ☐ Fornix ☐ Mons pubis ☐ Labia majora ☐ Labia minora ☐ Clitoris ☐ Urethral orifice
☐ Vestibule ☐ Perineum

☐ The pelvic vessels

☐ Internal iliac artery ☐ External iliac artery ☐ Medial umbilical ligament ☐ Internal iliac vein ☐ External iliac vein

The Endocrine System

☐ Locating the glands

☐ Hypothalamus ☐ Pituitary gland ☐ Pineal gland ☐ Thyroid gland ☐ Parathyroid glands ☐ Thymus
☐ Adrenal (suprarenal) glands ☐ Pancreas ☐ Ovaries ☐ Testes

☐ The pituitary gland

☐ Hypophysis ☐ Sella turcica ☐ Infundibulum ☐ Anterior lobe (adenohypophysis) ☐ Posterior lobe (neurohypophysis)

☐ The thyroid and parathyroid glands

☐ Thyroid gland, right lobe ☐ Thyroid gland, left lobe ☐ Isthmus ☐ Parathyroid glands ☐ Trachea ☐ Larynx

☐ Thyroid tissue, on the slide

FUNCTION ______________________ LOCATION ______________________

☐ Thyroid follicles ☐ Colloid ☐ Follicular cells ☐ Parafollicular (C) cells

☐ The adrenal gland

FUNCTION ______________________ LOCATION ______________________

☐ Adrenal (suprarenal) gland ☐ Adrenal cortex ☐ Zona glomerulosa ☐ Zona fasciculata ☐ Zona reticularis
☐ Adrenal medulla ☐ Kidney

☐ The endocrine pancreas

☐ Pancreas ☐ Pancreatic islets ☐ Acini

Vessels of the Abdomen and Pelvis

☐ Arteries of the abdomen

☐ Abdominal aorta ☐ Celiac trunk ☐ Left gastric artery ☐ Splenic artery ☐ Common hepatic artery
☐ Superior mesenteric artery ☐ Renal arteries ☐ Gonadal (testicular or ovarian) arteries ☐ Inferior mesenteric artery
☐ Right common iliac artery ☐ Left common iliac artery

☐ Arteries of the pelvis

☐ Internal iliac artery ☐ External iliac artery ☐ Medial umbilical ligament

☐ Veins of the abdomen and pelvis

☐ Inferior vena cava ☐ Renal veins ☐ Gonadal (testicular or ovarian) veins ☐ Right common iliac vein
☐ Left common iliac vein ☐ Internal iliac vein ☐ External iliac vein

☐ The hepatic portal system

☐ Hepatic portal vein ☐ Splenic vein ☐ Superior mesenteric vein ☐ Inferior mesenteric vein ☐ Hepatic veins

MODULE 5

Urinary and renal, reproductive, and the nervous system

64 competencies. TBL 8 Renal. TBL 9 Cranial nerves.

Weeks **14 to 17** Exam **5, Wed Dec 9** Lab practical and lecture exam **same day**

Week 14 Exam 4, then renal anatomy

MODULES 4 AND 5

BEFORE MONDAY, NOV 16

- [ ] Run a gap check on both banks and work the three day plan it builds
  Mastery OS
- [ ] Produce five brain dump prompts from this module on blank paper
  Brain dump practice
- [ ] Walk the lab structure list for this module and tick what you can name blind
  Lab sprints
- [ ] Sit a practice exam before exam day
  Practice Exam Builder

MON NOV 16 · EXAM DAY

- [ ] Lab practical
- [ ] Lecture exam

BEFORE WEDNESDAY, NOV 18

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for Renal Anatomy
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Bring this workbook

WED NOV 18 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Exam 4 rebuttals + Guided Renal Map
- [ ] Lab: Renal Anatomy
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind
  Brain dump practice

Competency loop, week 14

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Kidney gross anatomy and layers Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Internal kidney regions Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Nephron structure Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Cortical vs juxtamedullary nephrons Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
JGA and filtration membrane Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Kidney blood supply pathway Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Ureters, bladder, urethra and urine path Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______
Clinical anatomy of urinary disorders Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 14

You should need me less

Early in the term I tell you exactly where everything is. Course tools, then the group, then the tool. By now those directions have shortened, and soon they will just say to use the right retrieval tool.

That is not me withdrawing support. It is the point of the whole design. A student who can only work when someone tells them what to open next has learned this module. A student who can look at their own gaps and choose the tool that fixes them has learned how to learn anatomy, and that transfers to physiology, pathophysiology, pharmacology and everything after.

Ask yourself where you are on that line. When you sit down to study, do you know what you are weakest at right now? Do you know which tool addresses it? Can you tell the difference between a recall problem and an application problem in your own work?

If those questions have answers, you are close to independent. If they do not, that is the thing to work on for the rest of the term, and it is worth more than any single system you will be tested on.

Keep asking for help. Just aim to need a different kind of it.

Week 15 Renal review and reproductive anatomy

MODULE 5

BEFORE MONDAY, NOV 23

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for Male and Female Reproductive Anatomy
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Bring this workbook

MON NOV 23 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Kahoot: Renal
- [ ] Lab: Male and Female Reproductive Anatomy
- [ ] Round tables, practice rounds

WED NOV 25 · CAMPUS CLOSED

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind

Competency loop, week 15

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Scrotum and testis structure
Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Sperm duct system and pathway
Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Sperm cell structure
Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Accessory glands and penis
Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Clinical anatomy of male disorders
Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Ovary structure and follicle stages
Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Uterine tubes and ovum pathway
Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Uterus regions, ligaments, and wall
Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Vagina and external genitalia
Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Mammary gland structure
Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 15

What happens when the material stops being new

The last modules of this course cover systems you have partly met already. You have seen smooth muscle. You have seen simple squamous epithelium. You have traced vessels. That familiarity is useful and it is also a trap.

Familiar material invites you to skim. It looks like review, so you give it review effort, and then the exam asks you to distinguish two things you assumed you already had straight.

There is a specific move that helps here. Before you study a system that reuses old material, write down what you already believe about the reused part, from memory, before you read anything. Then read. The parts where your memory and the material disagree are exactly where your familiarity was hiding a gap.

The other risk this late in a term is drift. Everything from Module 1 is still testable on the cumulative final, and it has been sitting untouched for months. Keep at least a little of every study session pointed backwards. Two old prompts before you start the new ones is enough to keep the earlier modules alive rather than letting them go and rebuilding them in December.

Week 16 Renal TBL, brain, meninges and CSF

MODULE 5

BEFORE MONDAY, NOV 30

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for CNS: Brain, Meninges and CSF
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON NOV 30 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: CNS: Brain, Meninges and CSF
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, DEC 2

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for Brain Stem (CNS) and Cranial Nerves (PNS)
  Lab sprints

- [ ] Run the recall cards that are due
  Recall cards
- [ ] Bring this workbook

WED DEC 2 · IN CLASS AND LAB

- [ ] Brain dump. Round 1 retrieve, everything closed
- [ ] Clinical: Cranial Nerve Stations
- [ ] Lab: Brain Stem (CNS) and Cranial Nerves (PNS)
- [ ] Round tables, practice rounds

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind
  Brain dump practice

Competency loop, week 16

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Nervous tissue identification
Identify neurons and neuroglia in nervous tissue and the parts of a multipolar neuron.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Brain gross organization
Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cerebral surface and lobes
Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Functional cortical areas
Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cerebral white matter tracts
Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Basal ganglia
Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Diencephalon
Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cerebellum
Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Limbic system

Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cerebral arterial supply

Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Brainstem regions

Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Midbrain structures

Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Pons structures

Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Medulla oblongata structures

Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cranial nerve nuclei distribution

Match cranial nerve nuclei III through XII to the brainstem region that houses them.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Reticular formation

Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cranial meninges and dural reflections

Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Spinal meninges and meningeal spaces

Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Cord termination and lumbar puncture

Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Ventricular system

Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

CSF production and circulation

Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Blood-brain barrier structure

Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 HOOK ______

Nervous system divisions

Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Neuron structure

Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Synapse structures

Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Neuron classification

Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Neuroglia identification

Name the four CNS glia and two PNS glia and describe the support role each one performs.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Myelination and regeneration

Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

Cranial nerves

Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ____________

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

____

Something that still needs work

____

My priority for spaced retrieval next week

____

LEARNING TO LEARN · WEEK 16

What can you still retrieve from week 2?

Try it right now, before you read the rest of this. Blank paper. Everything closed. Reconstruct the epithelial classification system. Then list the four primary tissues with one defining feature each.

Whatever came out is your real answer to how well you learned Module 1, not the score you got on Exam 1 back in September. Those are different measurements and only one of them predicts how the cumulative final is going to go.

If it came out mostly intact, you have been coming back to it, and that is the system working. If it came out thin, you have found your December priority, and you found it with enough time to fix it.

This is worth doing for every module, one at a time, over the next couple of weeks. Do not reread first. Produce first, then check, then note what to work on. It takes about fifteen minutes per module and it will point you at the right material far better than a general sense of what feels shaky.

What you can still retrieve months later is the only version of knowing that counts.

Week 17 Cranial nerves, spinal cord and Exam 5

MODULE 5

BEFORE MONDAY, DEC 7

- [ ] Work the pre-work sheet with your notes packet and the reading open beside you
  Course materials
- [ ] Then watch the concept video and add to the same sheet in a second colour
  Course materials
- [ ] Pre-study the lab structures for CNS: Spinal Cord and Spinal Nerves
  Lab sprints
- [ ] Run the recall cards that are due
  Recall cards
- [ ] Second pass on the material for the TBL. Come able to defend an answer
- [ ] Bring this workbook

MON DEC 7 · TBL DAY

- [ ] iRAT
- [ ] tRAT, then appeals
- [ ] Application exercise with your team
- [ ] Lab: CNS: Spinal Cord and Spinal Nerves
- [ ] Round tables, practice rounds

BEFORE WEDNESDAY, DEC 9

- [ ] Run a gap check on both banks and work the three day plan it builds
  Mastery OS
- [ ] Produce five brain dump prompts from this module on blank paper
  Brain dump practice
- [ ] Walk the lab structure list for this module and tick what you can name blind

Lab sprints

- [ ] Sit a practice exam before exam day
  Practice Exam Builder

WED DEC 9 · EXAM DAY

- [ ] Lab practical
- [ ] Lecture exam

AFTER CLASS, AND ON YOUR NON-CLASS DAYS

- [ ] Mastery OS and recall cards for the lecture material
  Mastery OS
- [ ] Loops for the lab material
  Loops
- [ ] One prompt from an earlier module, produced blind
  Brain dump practice

Competency loop, week 17

1 name it · 2 note sheet · 3 video, second colour · 4 blind recall · 5 memory hook

Competency	Steps	Hook
CNS and PNS tissue terms Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Gray and white matter Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Reflex arc components Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Types of reflexes Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Neuronal pools and pathways Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Spinal cord external anatomy Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Roots, rootlets, and root ganglion Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Spinal cord internal anatomy Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______
Nerve connective tissue wrappings Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.	☐1 ☐2 ☐3 ☐4 ☐5	HOOK ______

Spinal nerves and rami

State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Sensory receptor classification

Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Nerve plexuses

Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Autonomic two-neuron pathway

Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

Autonomic ganglia and divisions

Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

☐1 ☐2 ☐3 ☐4 ☐5 HOOK ______

MY LEARNING THIS WEEK

Something I can do now that I could not do last week

______

Something that still needs work

______

My priority for spaced retrieval next week

______

LEARNING TO LEARN · WEEK 17

You built a learning system

Look back at your first few weeks in this book. The pages where you could barely produce anything on the brain dump. The drawings that were missing half their structures. The weeks where the reflection line said you did not know where to start.

Now look at what you can do. You can take a competency you have never seen, work it against the notes, check it against a video, produce it blind and build yourself a way to remember it. You can tell the difference between not recalling something and not being able to apply it. You can look at your own gaps and pick the tool that fixes them.

The anatomy matters and you will use it. But the anatomy is also the vehicle. What you actually built this term is a repeatable way of learning hard material, and no one can take that off you at the end of the semester.

Take this book with you. The loop on every week page works on physiology, on pharmacology, on anything with a lot of structure and a lot of detail. You already know it works, because you have the evidence in your own handwriting.

Lab structures Module 5

Tick the block name once you can identify it on the slide or the model, fill in function and location where the lines are there, then tick each structure you can point to. Anything not on this list is context, not something you will be asked to identify. The reference tables, regional and directional terms, quadrants and the rest, are in the standalone Module 5 structure list on the course site.

Nervous Tissue, Histology

☐ Multipolar neuron, on the slide and the model

FUNCTION ______________________ **LOCATION** ______________________

☐ Cell body (soma) ☐ Nucleus ☐ Nucleolus ☐ Cell processes (dendrites) ☐ Axon ☐ Axon hillock
☐ Synaptic bulb (terminal end) ☐ Nissl bodies ☐ Neurofibrils

☐ Spinal cord, cross section

FUNCTION ______________________ **LOCATION** ______________________

☐ White matter ☐ Gray matter ☐ Ventral (anterior) horn ☐ Dorsal (posterior) horn ☐ Lateral horn, if present
☐ Central canal

☐ Cerebral cortex

FUNCTION ______________________ **LOCATION** ______________________

☐ Pyramidal cells ☐ Gray matter of the cortex ☐ Underlying white matter

☐ Cerebellum

FUNCTION ______________________ **LOCATION** ______________________

☐ Purkinje cells ☐ Cerebellar cortex ☐ Arbor vitae

☐ Nerve, cross section

FUNCTION ______________________ **LOCATION** ______________________

☐ Axon ☐ Schwann cells ☐ Nodes of Ranvier ☐ Endoneurium ☐ Perineurium ☐ Epineurium ☐ Fascicle

☐ The neuroglia

☐ Astrocytes ☐ Oligodendrocytes ☐ Microglia ☐ Ependymal cells ☐ Schwann cells ☐ Satellite cells

The Brain

☐ Surface features of the cerebrum

☐ Gyrus ☐ Sulcus ☐ Longitudinal fissure ☐ Transverse fissure ☐ Precentral gyrus ☐ Precentral sulcus
☐ Central sulcus ☐ Postcentral gyrus ☐ Postcentral sulcus ☐ Lateral sulcus (Sylvian fissure) ☐ Insula

☐ The lobes

☐ Frontal lobe ☐ Parietal lobe ☐ Temporal lobe ☐ Occipital lobe ☐ Insula

☐ Midsagittal section

☐ Corpus callosum ☐ Sulcus of the corpus callosum ☐ Cingulate gyrus ☐ Cingulate sulcus ☐ Septum pellucidum
☐ Fornix ☐ Thalamus ☐ Massa intermedia (interthalamic adhesion) ☐ Hypothalamus ☐ Infundibulum ☐ Pineal gland
☐ Optic chiasm ☐ Cerebellum ☐ Transverse fissure

- [ ] **Inferior surface**

- [ ] Uncus - [ ] Parahippocampal gyrus - [ ] Mammillary bodies - [ ] Crus cerebri - [ ] Optic chiasm - [ ] Optic nerve - [ ] Optic tract - [ ] Olfactory bulb - [ ] Olfactory tract

- [ ] **Functional areas of the cortex**

- [ ] Primary motor cortex - [ ] Primary sensory cortex - [ ] Broca's area - [ ] Wernicke's area - [ ] Thalamus

- [ ] **The cerebellum**

- [ ] Cerebellar hemispheres - [ ] Vermis - [ ] Folia - [ ] Cerebellar cortex - [ ] Arbor vitae - [ ] Transverse fissure

- [ ] **The cerebral arterial circle (Circle of Willis)**

- [ ] Internal carotid artery - [ ] Anterior cerebral artery - [ ] Anterior communicating artery - [ ] Middle cerebral artery - [ ] Posterior communicating artery - [ ] Posterior cerebral artery - [ ] Basilar artery - [ ] Pontine arteries - [ ] Vertebral arteries - [ ] Superior cerebellar artery - [ ] Anterior inferior cerebellar artery - [ ] Posterior inferior cerebellar artery - [ ] Anterior spinal artery

The Brainstem and Cerebellum

- [ ] **The three regions**

- [ ] Midbrain - [ ] Pons - [ ] Medulla oblongata - [ ] Pontomedullary junction

- [ ] **The midbrain**

- [ ] Cerebral peduncles - [ ] Crus cerebri - [ ] Cerebral aqueduct

- [ ] **The pons**

- [ ] Basilar pons - [ ] Middle cerebellar peduncle - [ ] Fourth ventricle

- [ ] **The medulla oblongata**

- [ ] Pyramids - [ ] Decussation of the pyramids - [ ] Medullary olive - [ ] Pontomedullary junction

- [ ] **The cerebellum and its peduncles**

- [ ] Cerebellar hemispheres - [ ] Vermis - [ ] Folia - [ ] Arbor vitae - [ ] Superior cerebellar peduncle - [ ] Middle cerebellar peduncle - [ ] Inferior cerebellar peduncle

Meninges, Ventricles, and CSF

- [ ] **The cranial meninges**

- [ ] Dura mater - [ ] Periosteal layer of the dura - [ ] Meningeal layer of the dura - [ ] Arachnoid mater - [ ] Pia mater

- [ ] **The dural extensions**

- [ ] Falx cerebri - [ ] Falx cerebelli - [ ] Tentorium cerebelli

- [ ] **The dural venous sinuses**

- [ ] Superior sagittal sinus - [ ] Inferior sagittal sinus - [ ] Straight sinus - [ ] Confluence of sinuses - [ ] Transverse sinus - [ ] Sigmoid sinus

- [ ] **The meningeal spaces**

- [ ] Epidural space - [ ] Subdural space - [ ] Subarachnoid space - [ ] Arachnoid granulations

- [ ] **The ventricles**

- [ ] Lateral ventricle - [ ] Interventricular foramen of Monro - [ ] Third ventricle - [ ] Cerebral aqueduct - [ ] Fourth ventricle - [ ] Central canal - [ ] Septum pellucidum

- [ ] **The choroid plexus, on the slide and the model**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Choroid plexus - [ ] Ependymal cells - [ ] Capillary network

The Spinal Cord

- [ ] **The spinal meninges**

- [ ] Dura mater - [ ] Arachnoid mater - [ ] Pia mater - [ ] Epidural space - [ ] Subdural space - [ ] Subarachnoid space
- [ ] Denticulate ligaments

- [ ] **External anatomy of the cord**

- [ ] Cervical enlargement - [ ] Lumbar enlargement - [ ] Conus medullaris - [ ] Cauda equina - [ ] Filum terminale
- [ ] Anterior median fissure - [ ] Posterior median sulcus

- [ ] **The cord in cross section**

FUNCTION ______________________ **LOCATION** ______________________

- [ ] Ventral (anterior) horn - [ ] Lateral horn - [ ] Dorsal (posterior) horn - [ ] Gray matter - [ ] White matter - [ ] Gray commissure
- [ ] Central canal - [ ] Anterior white commissure - [ ] Anterior column - [ ] Lateral column - [ ] Posterior column

The Cranial Nerves

- [ ] **The twelve cranial nerves, I to XII**

- [ ] CN I, Olfactory - [ ] CN II, Optic - [ ] CN III, Oculomotor - [ ] CN IV, Trochlear - [ ] CN V, Trigeminal - [ ] CN VI, Abducens
- [ ] CN VII, Facial - [ ] CN VIII, Vestibulocochlear - [ ] CN IX, Glossopharyngeal - [ ] CN X, Vagus - [ ] CN XI, Accessory
- [ ] CN XII, Hypoglossal

- [ ] **Structures associated with the cranial nerves**

- [ ] Pontomedullary junction - [ ] Internal acoustic meatus - [ ] Olfactory bulb - [ ] Olfactory tract
- [ ] Olfactory foramina in the cribriform plate - [ ] Optic nerve - [ ] Optic chiasm - [ ] Optic tract - [ ] Superior orbital fissure
- [ ] Optic canal

- [ ] **The branches of the trigeminal nerve**

- [ ] Ophthalmic division (V1) - [ ] Maxillary division (V2) - [ ] Mandibular division (V3)

- [ ] **The extrinsic eye muscles**

- [ ] Lateral rectus - [ ] Medial rectus - [ ] Superior rectus - [ ] Inferior rectus - [ ] Superior oblique - [ ] Inferior oblique
- [ ] Levator palpebrae superioris

Peripheral Nerves and Plexuses

- [ ] **The spinal nerve and its roots**

- [ ] Ventral (anterior) root - [ ] Dorsal (posterior) root - [ ] Dorsal root ganglion - [ ] Spinal nerve - [ ] Intervertebral foramen
- [ ] Rootlets

- [ ] **The rami and the sympathetic connections**

- [ ] Ventral (anterior) ramus - [ ] Dorsal (posterior) ramus - [ ] Gray ramus communicans - [ ] White ramus communicans
- [ ] Sympathetic chain ganglion - [ ] Sympathetic trunk

- [ ] **The plexuses**

- [ ] Cervical plexus - [ ] Brachial plexus - [ ] Lumbar plexus - [ ] Sacral plexus

- [ ] **Nerves to find on the cadaver and the models**

- [ ] Phrenic nerve
- [ ] Intercostal nerves

PART C

Brain dump practice bank

This is the content practice bank. Every prompt here is anatomy, and every one of them is fair game. Use it as a menu. Pick five at random on a study night, set a clock, produce what you can on blank paper, then check yourself against your notes, the Atlas and the lab sprints. Nothing here is an answer sheet, and there is no key at the back. Producing the answer is the practice.

Module 1 · Foundations, cells, tissues and the integument

INTRO TO ANATOMY

- [ ] **Directional terms, paired.** List every directional term you can and write its opposite beside it. Then draw a stick figure in anatomical position and mark four of your pairs on it.
- [ ] **Planes and abdominal divisions.** Draw the three anatomical planes on a body outline. Name each one and write what view it gives you. Then draw the four abdominal quadrants and name them.
- [ ] **Cavities from the outside in.** Dump the body cavities from the outside in. Name each cavity, what it holds, and which membrane lines it.
- [ ] **Regional terms mind map.** Draw a body outline. Put anterior regional terms down the left side and posterior terms down the right, with a line from each name to the spot it points to.
- [ ] **Levels of organization.** Write the levels of structural organization from smallest to largest and give one anatomy example at each level.
- [ ] **Serous membranes.** Draw a fist pushed into a balloon. Label the parietal layer, the visceral layer, and the cavity. Then name the three serous membranes and the organ each one wraps.
- [ ] **Nine regions, filled in.** Draw the nine abdominopelvic regions in their grid, name every one, and write one organ that sits in each.
- [ ] **Behind the peritoneum.** List every retroperitoneal structure you can. Then write two organs that are intraperitoneal and say what the difference means for where they sit.
- [ ] **The mediastinum.** Draw the thorax in cross section. Mark the mediastinum, name what forms its four borders, and list everything inside it.
- [ ] **What each cut shows.** Pick the kidney. Sketch what a sagittal, a frontal, and a transverse cut through it would each show. Then name three surface landmarks you can find on yourself.

CELL ANATOMY

- [ ] **Organelle inventory.** List every organelle you can name and write one job beside each one. Do not stop to make it pretty.
- [ ] **Draw the cell.** Draw a cell and label everything you can from memory. Then circle the three structures a protein passes through on its way out of the cell.
- [ ] **Protein route, in order.** Write the route a protein takes from the gene to the outside of the cell, naming every structure it passes through in order. Then list every membrane-bound organelle you can.
- [ ] **Nucleus at the center.** Draw a mind map with the nucleus in the middle. Branch out to every structure it directs or supplies, with an arrow on each branch showing which way the material moves.
- [ ] **The nucleus in detail.** Draw the nucleus enlarged and label every part. Then write what the nuclear envelope is continuous with.

- [ ] **The endomembrane system.** Name every organelle in the endomembrane system and draw arrows showing how material moves from one to the next.
- [ ] **Inside a mitochondrion.** Draw a mitochondrion cut open and label everything. Then write two features that make it unlike any other organelle.
- [ ] **Cell junctions.** Name the three main cell junctions. For each, draw a quick sketch of what it looks like between two cells and write where in the body you would find it.
- [ ] **Organelle or inclusion.** Split a page in two. On the left list organelles, on the right list cytoplasmic inclusions. Write what makes something an inclusion rather than an organelle.
- [ ] **Mitosis in order.** Name the stages of mitosis in order and draw a quick sketch of the chromosomes at each stage.

CELL ANATOMY

- [ ] **The plasma membrane.** Dump everything you know about the plasma membrane: what it is built from, what sits in it, and what can cross it without help.
- [ ] **Cytoskeleton and cell surface.** List the three cytoskeletal elements with one job each. Then name every motile or surface projection you can and write where you would find it.
- [ ] **Tell them apart.** One line each, no notes: rough ER, smooth ER, Golgi, lysosome, peroxisome, mitochondrion. Say what makes each one different from the others.
- [ ] **Membrane proteins by job.** List the jobs membrane proteins do and draw a quick example of each sitting in a bilayer.
- [ ] **Three surface extensions.** Draw microvilli, cilia, and a flagellum side by side on three cells. Label each, write its internal structure, and name a location.
- [ ] **Free versus bound ribosomes.** Draw a cell and mark where free ribosomes sit and where bound ribosomes sit. Write what each group makes and where that product ends up.
- [ ] **A lysosome from start to finish.** Draw the life of a lysosome: where it is made, what it fuses with, and what happens to the contents.
- [ ] **Centrosome and spindle.** Draw a centrosome and label its parts. Then sketch what it does during cell division.
- [ ] **Break one thing.** Pick one organelle and build a mind map of what would go wrong in the cell if it stopped working. Branch out at least three consequences.
- [ ] **What the scope can show.** Draw two columns, light microscope and electron microscope. Sort as many cell structures as you can into the column where you could actually see them.

BASIC HISTOLOGY AND TISSUES

- [ ] **Epithelium by name.** List every epithelial type you can, named by layering and cell shape, and give one location for each.
- [ ] **What connective tissue is made of.** Dump the three ingredients of every connective tissue and name what falls under each. Then name three connective tissue proper subtypes.
- [ ] **Simple columnar versus pseudostratified.** Draw a simple columnar and a pseudostratified columnar epithelium side by side and label both. Then write how you tell them apart under the scope.
- [ ] **Epithelium mind map.** Build a mind map with epithelium at the center. One main branch for layering, one for cell shape. Fill in every combination that actually exists and hang a location off each.
- [ ] **Polarity and the basement membrane.** Draw an epithelial sheet and label the apical surface, the lateral surfaces, the basal surface, and the basement membrane. Write what sits on the other side of the basement membrane.
- [ ] **Glands, sorted.** Split glands two ways: exocrine versus endocrine, then unicellular versus multicellular. Give an example in every box you can fill.
- [ ] **Two kinds of stratified squamous.** Draw keratinized and non-keratinized stratified squamous side by side. Label both and write where each one is found.
- [ ] **Areolar tissue.** Draw areolar connective tissue as it looks on a slide and label everything in it. Then write why it is called the packing tissue.
- [ ] **Adipose and reticular.** Draw adipose tissue and reticular tissue side by side. Label the cells and fibers in each and name two locations for each.

- [ ] **Dense regular versus irregular.** Draw dense regular and dense irregular connective tissue. Show the fiber direction in each, and write what that direction lets the tissue resist.

BASIC HISTOLOGY AND TISSUES

- [ ] **The four primary tissues.** Name the four primary tissue types. For each, write one defining feature and two locations.
- [ ] **Three muscle tissues.** List the three muscle tissue types. For each, write striations, nuclei per cell, branching, and whether it is voluntary.
- [ ] **Neuron and neuroglia.** Draw a neuron, label every part you can, and mark the direction the signal travels. Then name every glial cell you can with one job each.
- [ ] **Four tissues, one map.** Build a mind map with the four primary tissues as main branches. Hang every subtype you can under each branch.
- [ ] **Cardiac muscle up close.** Draw cardiac muscle as it looks on a slide and label everything that identifies it. Write the two features that no other tissue has.
- [ ] **Body membranes.** Name the four body membranes. For each write which tissues build it and where it is found.
- [ ] **Name that slide.** For each clue write the tissue: lines the trachea and looks layered but is not. Has lacunae in a glassy matrix. Has a central canal ringed by circles. Has one central nucleus and no stripes. Has fibers going in every direction under the skin.
- [ ] **Skeletal muscle on a slide.** Draw skeletal muscle in longitudinal section and label what you can see. Then write where the nuclei sit and why that matters for identifying it.
- [ ] **Myelin, two ways.** Draw a myelinated axon in the CNS and one in the PNS. Label the cell doing the wrapping in each and write one difference between them.
- [ ] **Four tissues in one organ.** Draw the wall of the gut tube in cross section and label where each of the four primary tissues appears in it.

INTEGUMENTARY

- [ ] **Epidermis, deep to superficial.** Write the epidermal layers from deepest to most superficial and one thing that happens in each. Note which layer is not in thin skin.
- [ ] **Skin in section.** Draw skin in section from the surface down to the hypodermis and label every structure you can.
- [ ] **Glands and appendages.** List the skin glands and appendages. For each write what it secretes and where the secretion goes. Then write the difference between holocrine, merocrine, and apocrine.
- [ ] **Cells of the epidermis.** Name the four cell types in the epidermis. For each, write its job and which layer you would find it in.
- [ ] **The dermis.** Draw the dermis and split it into its two layers. Label what is in each and write which one gives you fingerprints.
- [ ] **A hair and its follicle.** Draw a hair in its follicle from the bulb up past the skin surface. Label every part you can.
- [ ] **A nail.** Draw a nail from above and from the side. Label every named part.
- [ ] **Skin color and burn depth.** Name the three pigments that contribute to skin color. Then draw skin in section and mark how deep a first, second, and third degree burn reaches.
- [ ] **Function to structure.** Build a mind map with skin at the center. Branch to each function it performs and, on the end of every branch, name the structure that actually does it.
- [ ] **Receptors by depth.** Draw skin in section and place every sensory receptor you can at the depth where it actually sits. Write what each one detects.

Module 2 · Bone tissue, axial and appendicular skeleton, joints

BONE TISSUE

- [ ] **Long bone in section.** Draw a long bone cut lengthwise and label every region and covering you can.
- [ ] **The osteon, center out.** Dump the osteon starting at the center and working out. Name every structure and say what runs through it.

- [ ] **Bone cells, compact versus spongy.** List the four bone cell types with what each one does. Then write how compact bone differs from spongy bone in structure and in where you find it.
- [ ] **Long bone microanatomy, labeled.** Draw a wedge of compact bone at the tissue level and label every structure in it. Then add a patch of spongy bone beside it and label that too.
- [ ] **What bone is made of.** Split bone matrix into its organic and inorganic parts. Name what is in each and write what property each part gives the bone.
- [ ] **Periosteum and endosteum.** Draw the surface of a long bone enlarged. Label both coverings, their layers, and what anchors the outer one to the bone.
- [ ] **Bones by shape.** Name the five bone shape classes. Give two examples of each and write one structural feature that goes with the shape.
- [ ] **Feeding a bone.** Draw a long bone and trace the blood and nerve supply into it. Name every vessel and canal along the way.
- [ ] **Bone as an organ.** Build a mind map with one long bone at the center. Branch to every tissue type present in it and name where each one sits.
- [ ] **Two ways to build bone.** Draw intramembranous and endochondral ossification side by side. Write what each one starts from and name two bones formed by each.

AXIAL SKELETON

- [ ] **Cranial then facial.** List the cranial bones, then list the facial bones. Mark which ones are paired.
- [ ] **Skull, lateral view.** Draw the skull from the side. Label every bone you can and every suture you can.
- [ ] **Openings and what goes through.** Name every skull foramen or opening you can and write what passes through it. Then list the bones that build the orbit.
- [ ] **The cranial floor.** Draw the inside of the cranial base from above. Outline the three cranial fossae, name each, write which bones form it, and name one thing that sits in it.
- [ ] **The sphenoid alone.** Draw the sphenoid bone by itself and label every part and opening on it.
- [ ] **The ethmoid alone.** Draw the ethmoid bone and label its parts. Then write which three cavities it helps build.
- [ ] **Mandible and the jaw joint.** Draw the mandible and label every named landmark. Then mark where it articulates with the skull and name that joint.
- [ ] **The bone that touches nothing.** Draw the hyoid bone, label its parts, and write where it sits and what attaches to it.
- [ ] **Sutures and soft spots.** Draw the skull from above and from the side. Label every suture. Then name the fontanelles of a newborn skull and say why they exist.
- [ ] **What the skull is for.** Build a mind map with the skull at the center. Branch to each job it does and name the specific structures that do it.

AXIAL SKELETON, SPINE AND THORACIC CAGE

- [ ] **Counts and giveaways.** Write the vertebral counts by region, top to bottom. Then write one feature that tells you a vertebra came from each region.
- [ ] **Typical vertebra from above.** Draw a typical vertebra viewed from above and label every part. Then mark what you would add to make it cervical.
- [ ] **The thoracic cage.** Dump the thoracic cage: parts of the sternum, the rib groups, and the parts of a typical rib.
- [ ] **The column from the side.** Draw the whole vertebral column from the side. Mark the four curvatures, name each, and label which are primary and which are secondary.
- [ ] **Atlas and axis.** Draw C1 and C2 from above, side by side. Label every part and write what movement each joint allows.
- [ ] **Sacrum and coccyx.** Draw the sacrum from the front and from the back. Label every named feature, then add the coccyx.
- [ ] **The disc.** Draw two vertebrae with the disc between them, cut in half. Label the parts of the disc and mark where a herniation usually goes and what it presses on.
- [ ] **Ligaments holding the spine.** Draw two or three vertebrae from the side and label every ligament you can place on them.

- [ ] **Region by region.** Build a mind map with the vertebral column at the center. One branch per region. On each branch put the count, the identifying feature, and the movement that region does best.
- [ ] **Sternum and rib joints.** Draw the sternum and label its three parts and its landmarks. Then name the joints a typical rib makes at both ends.

BONE TISSUE AND CARTILAGE

- [ ] **Three cartilages.** List the three cartilage types. For each write what is in the matrix, what it looks like under the scope, and two locations.
- [ ] **How a long bone gets longer.** Write how a long bone grows in length. Name the zones of the growth plate in order and say what happens in each.
- [ ] **Bone markings.** Name as many bone markings as you can, sorted into projections, depressions, and openings. Give an example bone for each.
- [ ] **Wider versus longer.** Draw a bone growing in width beside one growing in length. Name the process in each and say which cells are working on which surface.
- [ ] **Remodeling and repair.** Name the cells that remodel bone and what each does. Then draw the stages of fracture repair in order.
- [ ] **Feeding cartilage.** Draw a piece of hyaline cartilage with its covering. Label the covering, the cells, and the matrix, then write how nutrients reach the chondrocytes.
- [ ] **The scapula.** Draw the scapula from the back and from the side. Label every named feature.
- [ ] **The humerus.** Draw the humerus from the front and from the back. Label every landmark on both sides.
- [ ] **Radius and ulna.** Draw the radius and ulna in anatomical position, label both fully, and show how they articulate with each other and with the humerus.
- [ ] **Cartilage mind map.** Build a mind map with cartilage at the center. One branch per type. On each branch put the matrix fibers, the appearance on a slide, and two locations.

APPENDICULAR SKELETON

- [ ] **Upper limb, shoulder to fingertips.** List the bones of the upper limb from the shoulder to the fingertips, including the carpals in their two rows.
- [ ] **Lower limb, hip to toes.** List the bones of the lower limb from the hip to the toes, including the tarsals.
- [ ] **The pelvis.** Draw the pelvis and label every part you can. Then write two differences between a male and a female pelvis.
- [ ] **The femur.** Draw the femur from the front and from the back and label every named landmark.
- [ ] **Tibia and fibula.** Draw the tibia and fibula together and label both. Mark which one bears weight and which one does not.
- [ ] **Arches of the foot.** Draw the foot from the side and from below. Mark the three arches, name them, and draw arrows showing how weight moves from the heel to the toes.
- [ ] **The hand.** Draw the bones of the wrist and hand in position and label every one. Keep the carpals in their correct two rows.
- [ ] **Two girdles compared.** Draw the pectoral girdle and the pelvic girdle side by side. For each, write what it attaches to and whether it favors mobility or stability, and say what structure makes that true.
- [ ] **Appendicular mind map.** Build a mind map with the appendicular skeleton at the center. Branch to each girdle, then out through the limb bone by bone.
- [ ] **Find it on yourself.** List every bony landmark you can actually feel on your own body. Write the bone it belongs to and roughly where it is.

Module 3 · Cardiovascular, muscle and blood

HEART ANATOMY AND BLOOD VESSELS

- [ ] **Trace a drop of blood.** Draw the heart and trace one drop of blood from the vena cava all the way to the aorta. Name every chamber, valve, and vessel it passes in order.

- [ ] **Coronary supply and wall layers.** Name every coronary vessel you can and the region of the heart it feeds. Then write the layers of the heart wall from the inside out.
- [ ] **Conduction and the vessel wall.** Write the conduction pathway of the heart in order, naming each structure. Then write the three layers of a vessel wall and how an artery differs from a vein.
- [ ] **Pulmonary circulation.** Draw the pulmonary circuit on its own. Start in the right ventricle and finish in the left atrium, naming every vessel and structure on the way. Mark clearly where the blood picks up oxygen.
- [ ] **Off the aorta.** Draw the aorta from the left ventricle down to where it splits. Name each region and every major branch coming off it.
- [ ] **The pericardium.** Draw the heart inside its sac, cut in section. Label every layer from the outside in and name the space that holds the fluid.
- [ ] **Why the walls differ.** Draw the four heart chambers in cross section with the wall thickness roughly to scale. Write what each chamber pumps to and why its wall is the thickness it is.
- [ ] **Three kinds of capillary.** Name the three capillary types. For each, draw the wall, describe what makes it leaky or tight, and name a place you would find it.
- [ ] **Heart mind map.** Build a mind map with the heart at the center. Four branches: chambers, valves, vessels, conduction. Fill in every branch fully.
- [ ] **Coming back to the heart.** Draw the major veins returning blood to the right atrium. Name each one and the region it drains.

MUSCLE STRUCTURE AND SARCOMERES

- [ ] **Whole muscle down to filament.** Dump muscle from the whole organ down to the myofilament. Name every level and every wrapping in order.
- [ ] **Draw a sarcomere.** Draw one sarcomere and label every band, line, and zone. Then mark what gets shorter when the muscle contracts and what does not.
- [ ] **Filament proteins and the triad.** List the proteins of the thick filament and the thin filament and what each one does. Then write what a triad is and what it contains.
- [ ] **SR and T tubules.** Draw a stretch of a muscle fiber cut open. Label the sarcoplasmic reticulum, the terminal cisternae, and the T tubules, and show where the triad sits relative to the sarcomere.
- [ ] **The neuromuscular junction.** Draw the neuromuscular junction and label everything on both sides of the gap.
- [ ] **Fiber types.** Name the skeletal muscle fiber types and, for each, write the structural features that go with it: color, mitochondria, capillary supply, and fiber diameter.
- [ ] **How muscle attaches.** Draw a muscle spanning a joint and label how it attaches at both ends. Name the connective tissue continuity from one bone to the other.
- [ ] **Sliding filament map.** Build a mind map of what has to be in place for a sarcomere to shorten. Branch out to every structure involved and put them in order of use.
- [ ] **Bone to bone.** Draw a whole muscle in cross section and label every connective tissue layer from the outside in. Then write what all of those layers become at the end of the muscle.
- [ ] **The motor unit.** Draw one motor neuron supplying its muscle fibers. Write what a motor unit is and compare a small one with a large one.

HEART ANATOMY

- [ ] **The two circuits.** Draw the pulmonary and systemic circuits as one loop. Name the great vessels, mark which side is the high pressure side, and mark where the blood is oxygenated.
- [ ] **Valves, open and shut.** List the four heart valves, where each one sits, how many cusps it has, and when it is open versus closed.
- [ ] **Surface features and landmarks.** Draw the heart from the front and label every surface feature, sulcus, and external landmark you can. Then name the fetal shunts and the adult remnant of each.
- [ ] **The heart in the chest.** Draw the thorax from the front and place the heart in it accurately. Mark the base, the apex, the borders, and which ribs and spaces they line up with.

- [ ] **Where to listen.** Draw the chest wall and mark the four points where each heart valve is best heard. Write why the listening point is not directly over the valve.
- [ ] **Muscles of expression.** Draw a face and label every muscle of facial expression you can, with the action each one produces.
- [ ] **Muscles of chewing.** Name the four muscles of mastication. Draw each, mark its attachments, and write what movement it makes at the jaw.
- [ ] **The back, layer by layer.** Draw the back and label the superficial muscles. Then write which layer sits underneath and name one muscle in it.
- [ ] **Chest and thoracic wall.** Draw the anterior chest and label the muscles of the pectoral region and the thoracic wall, with the action of each.
- [ ] **Decode a muscle name.** Pick five muscles. For each, break the name apart and write what each piece tells you: shape, size, location, fiber direction, attachments, or action.

BLOOD

- [ ] **Formed elements.** List the formed elements of blood. For each write its job and roughly how common it is.
- [ ] **White cells under the scope.** Split the white cells into granulocytes and agranulocytes. Name each one and write the feature you would use to identify it on a slide.
- [ ] **Blood types and hemoglobin.** Write the ABO and Rh types with the antigen and antibody for each, then say who can receive from whom. Then name the parts of a hemoglobin molecule.
- [ ] **A blood smear.** Draw a blood smear as it looks under the scope. Include every formed element in roughly the right proportion and label each one.
- [ ] **Why a red cell is that shape.** Draw an erythrocyte from the side and from above. Label its shape, write what it lacks, and give two reasons the shape helps it work.
- [ ] **Nerves into the arm.** Draw the brachial plexus in simple form and name the five terminal branches. Then write which compartment of the arm each one supplies.
- [ ] **Anterior compartment.** Draw the anterior arm and anterior forearm. Label the muscles in each and write the main action of the group.
- [ ] **Arteries down the arm.** Trace the arterial supply from the aorta to the fingertips, naming every named artery in order and where it changes name.
- [ ] **Blood mind map.** Build a mind map with blood at the center. Two main branches, plasma and formed elements. Fill both out completely.
- [ ] **Red marrow, yellow marrow.** Draw a long bone and a flat bone and mark where red marrow and yellow marrow sit in each. Write how that distribution changes from child to adult.

BLOOD

- [ ] **Hemopoiesis.** Start at the hemopoietic stem cell and branch out as far as you can name. Get every cell line you can onto the page.
- [ ] **What is in plasma.** List everything in plasma you can name, grouped as proteins, electrolytes, nutrients, wastes, and gases.
- [ ] **Stopping a bleed.** Write the steps of hemostasis in order. Name the cells and proteins doing the work at each step.
- [ ] **Where lymph rejoins the blood.** Draw the two main lymphatic ducts and mark where each empties into the venous system. Then write which parts of the body each one drains.
- [ ] **Where old red cells go.** Name the organs that break down worn out red blood cells. For each, name the structure inside it that does the work and write what happens to the pieces.
- [ ] **Posterior compartment.** Draw the posterior arm and posterior forearm. Label the muscles and write the action of each group.
- [ ] **The radial nerve.** Trace the radial nerve from the plexus to the hand. Name where it wraps the humerus and write what is lost if it is damaged there.

- [ ] **The carpal tunnel.** Draw the wrist in cross section. Label the roof, the floor, and everything passing through the tunnel. Then name one structure that passes over the top and is therefore spared.
- [ ] **A red cell from birth to breakdown.** Build a mind map following one red blood cell from stem cell to disposal. Put every structure and organ it passes on the branches.
- [ ] **Veins of the upper limb.** Draw the arm and label the superficial and the deep veins. Mark where you would draw blood and say why that spot is used.

Module 4 · Respiratory, lymphatic, digestive and endocrine

RESPIRATORY ANATOMY AND HISTOLOGY

- [ ] **Nostril to alveolus.** Trace one molecule of air from the nostril to the alveolus. Name every structure it passes through in order.
- [ ] **Lungs and pleurae.** Draw both lungs with their lobes and fissures. Label the pleurae and their cavity, then write what sits in the space between the lungs.
- [ ] **Epithelium down the airway.** Write the epithelium at each level of the airway and note what changes as you go deeper. Then name the cells you find in an alveolus.
- [ ] **The blood side of the lung.** Draw the pulmonary circulation as it reaches the lung tissue. Show the artery arriving, the capillary bed around an alveolus, and the vein leaving. Mark where the blood changes color and why.
- [ ] **The respiratory membrane.** Draw the barrier between alveolar air and blood, enlarged. Label every layer the oxygen must cross and write roughly how thin it is.
- [ ] **Segments and the root.** Name the lobes of each lung and write how many bronchopulmonary segments each lung has. Then draw the root of the lung and label what enters and leaves at the hilum.
- [ ] **Inside the nose.** Draw the lateral wall of the nasal cavity and label everything on it. Mark where each sinus and the tear duct drain.
- [ ] **Three pharynges.** Draw the pharynx from the side and split it into its three parts. Label the boundaries of each and every opening into it.
- [ ] **Two zones.** Build a mind map splitting the airway into the conducting zone and the respiratory zone. Put every structure on the correct branch and mark where the line falls.
- [ ] **The diaphragm.** Draw the diaphragm from below. Label its attachments and the three major openings, and name what passes through each.

ENDOCRINE ANATOMY

- [ ] **Gland map.** List every endocrine gland or endocrine tissue you can and write where in the body it sits. Draw a rough body outline and place them on it.
- [ ] **Hypothalamus and pituitary.** Draw the hypothalamus and pituitary and label the parts, including how each lobe connects to the brain. Then list the hormones of the anterior and posterior lobes.
- [ ] **Four glands, region by region.** For the thyroid, parathyroid, adrenal, and pancreas: describe the gross structure, name the regions or layers inside it, and say what each region secretes.
- [ ] **The other endocrine organs.** Draw and place the pineal gland, the thymus, and the gonads. For each write where it sits, what it secretes, and how it changes with age.
- [ ] **Adrenal in section.** Draw an adrenal gland cut in half. Label the capsule, the three cortical zones in order, and the medulla, with what each region secretes.
- [ ] **Two pancreases on one slide.** Draw a pancreas slide showing both tissue types. Label the exocrine and endocrine parts and write what each one releases and where it goes.
- [ ] **Anterior thigh.** Draw the anterior thigh and label every muscle. Write the action of each and name the group they belong to.
- [ ] **The femoral triangle.** Draw the femoral triangle and label its three borders. Then name the contents from lateral to medial.

- [ ] **Hypothalamus at the center.** Build a mind map with the hypothalamus in the middle. Branch to every structure it controls and mark whether the control runs through a blood portal or down an axon.
- [ ] **Hidden endocrine tissue.** Name every organ that is not primarily endocrine but still contains hormone-secreting cells. Write the cells and where in the organ they sit.

RESPIRATORY

- [ ] **Muscles of breathing.** Name the muscles of quiet breathing and the muscles brought in for forced breathing. For each, say what it does to the size of the thoracic cavity.
- [ ] **The bronchial tree, in order.** Write the branching order of the bronchial tree from the trachea to the alveolus. Note where cartilage stops and where gas exchange starts.
- [ ] **Sinuses and larynx.** Name the paranasal sinuses and the bone each is in. Then name every laryngeal cartilage you can and the two sets of folds.
- [ ] **Trachea in cross section.** Draw the trachea cut across and label every layer and structure from the lumen outward.
- [ ] **Pleural recesses.** Draw the pleural cavity and mark the recesses. Name each one and write where fluid would collect in someone sitting upright.
- [ ] **Posterior thigh.** Draw the posterior thigh and label the hamstring group plus any other muscle you can place. Write the two actions the group shares and where they attach proximally.
- [ ] **The sciatic nerve.** Trace the sciatic nerve from where it forms to where it divides. Name its two divisions and write what each supplies.
- [ ] **Gluteal region.** Draw the gluteal region and label the muscles from superficial to deep. Then mark the safe quadrant for an intramuscular injection and say what you are avoiding.
- [ ] **One molecule of air.** Build a mind map following one molecule of air in and back out. Branch at every point where the pathway divides and note what changes at each level.
- [ ] **Landmarks on the chest.** Draw the chest wall from the front and back. Mark the surface landmarks you would use to count ribs, find the lung borders, and place a stethoscope.

GI TRACT OVERVIEW

- [ ] **Mouth to anus.** Trace a bite of food from the mouth to the anus. Name every organ and every sphincter it passes in order.
- [ ] **The four layers of the gut wall.** Draw the GI wall in cross section and name the four layers from the lumen outward. Write what is inside each layer.
- [ ] **Peritoneum and what sits behind it.** Name every peritoneal fold, mesentery, and omentum you can. Then list the retroperitoneal structures.
- [ ] **Inside the mouth.** Draw the open mouth and label everything you can. Then name the three pairs of salivary glands and mark where each duct opens.
- [ ] **Teeth.** Draw the adult dental arch and label the tooth types with how many of each. Then draw one tooth in section and label every layer.
- [ ] **Swallowing route.** Trace a bolus from the mouth to the stomach, naming every structure it passes. Mark the point where the airway has to be shut off and name what does it.
- [ ] **Two compartments of the leg.** Draw the lower leg in cross section. Label the anterior, lateral, and posterior compartments, name the muscles in each, and write the shared action of each group.
- [ ] **Feeding the gut.** Draw the abdominal aorta and its three unpaired branches. For each, name the gut region it supplies and write where one territory hands over to the next.
- [ ] **GI mind map.** Build a mind map with the GI tract at the center. One branch per organ in order. On each branch write one structural feature and one thing that happens there.
- [ ] **The gut has its own nerves.** Draw the gut wall and mark the two nerve plexuses. Name each, write which layer it sits in, and say what it controls.

GI REGIONAL ANATOMY

- [ ] **Liver and the bile route.** Draw the liver and label the lobes and the ligaments. Then trace bile from a hepatocyte to the duodenum, naming every duct.
- [ ] **The stomach.** Draw the stomach and label every region, curvature, and sphincter. Then list the cell types in a gastric gland and what each one secretes.
- [ ] **The small intestine.** Name the three regions of the small intestine in order with one identifying feature of each. Then name every structure that increases surface area, largest to smallest.
- [ ] **The large intestine.** Draw the large intestine and label every region in order. Then name the three features unique to it and mark them on your drawing.
- [ ] **The end of the line.** Draw the rectum and anal canal in section. Label every structure and both sphincters, and write which one you control.
- [ ] **The biliary tree.** Draw the whole biliary tree from the liver to the duodenum. Label every duct and the gallbladder, and mark where bile is stored and where it enters the gut.
- [ ] **The portal system.** Draw the hepatic portal system. Name the veins that form the portal vein and trace the blood through the liver and back to the heart.
- [ ] **Mesenteries and omenta.** Draw the abdomen in sagittal section and label every peritoneal fold. Name what each one connects and what runs inside it.
- [ ] **A fatty meal.** Build a mind map following a fatty meal. Branch to every organ that handles it and name the structure in each organ doing the work.
- [ ] **The right lower quadrant.** Draw the cecum, appendix, and ileocecal junction. Label every part and mark the surface point that overlies the appendix.

GI ACCESSORY ORGANS

- [ ] **The liver, gross to microscopic.** Draw the liver and label the lobes and ligaments. Then draw one lobule and label what runs in the portal triad.
- [ ] **Bile out, enzymes in.** Trace bile from the hepatocyte to the duodenum, naming every duct. Then name the parts of the gallbladder and the pancreatic ducts.
- [ ] **The pancreas in place.** Draw the pancreas and label its four regions. Then name what it sits against in front and behind.
- [ ] **Two blood supplies.** Draw the liver and show both of its blood supplies arriving. Name each vessel, write what it delivers, and give the rough share of total flow.
- [ ] **Portal triad at high power.** Draw one liver lobule and label everything. Zoom in on a corner and draw the portal triad with all three of its structures.
- [ ] **Where bile comes from and goes.** Write which cells make bile, where it is stored and concentrated, and what triggers its release. Draw the route as a line with the structures named along it.
- [ ] **Accessory organ mind map.** Build a mind map with the three accessory organs as branches. On each, name its duct, what it delivers, and exactly where that lands in the gut.
- [ ] **Where the ducts meet the gut.** Draw the second part of the duodenum opened up. Label the papillae, the ampulla, and the sphincter, and mark which duct arrives at each.
- [ ] **Behind or in front.** For the liver, gallbladder, and pancreas, write whether each is intraperitoneal or retroperitoneal, and name the peritoneal structure attached to it.

Module 5 · Urinary, reproductive and the nervous system

RENAL ANATOMY

- [ ] **Kidney in coronal section.** Draw a kidney cut front to back and label every region and structure. Then trace urine from a collecting duct to the outside of the body.
- [ ] **The nephron in order.** Name every segment of the nephron in order from the glomerular capsule to the collecting duct. Mark which parts sit in the cortex and which dip into the medulla.

- [ ] **Renal blood route and the corpuscle.** List the blood vessels in order from the renal artery to the renal vein. Then name the parts of the renal corpuscle and its filtration barrier.
- [ ] **The kidney in place.** Draw the kidney in position from behind. Mark which vertebral levels it spans, which one sits lower and why, and label every covering layer from the kidney outward.
- [ ] **The juxtaglomerular apparatus.** Draw the glomerulus with the tubule looping back to touch it. Label the two cell populations of the juxtaglomerular apparatus and write what each one senses.
- [ ] **Down the ureter.** Draw the ureter from the renal pelvis to the bladder. Mark its three narrow points and label the layers of its wall.
- [ ] **The bladder wall.** Draw the bladder in section and label its wall layers and its internal landmarks. Write what changes as it fills.
- [ ] **Two kinds of nephron.** Draw a cortical and a juxtamedullary nephron side by side in a wedge of kidney. Label the difference in loop length and in the capillaries around each.
- [ ] **Kidney mind map.** Build a mind map with the kidney at the center. Branch to filter, reabsorb, and drain. Put the exact structure that does each job on the branch.
- [ ] **From papilla to pelvis.** Draw the collecting system on its own, stripped of the rest of the kidney. Name every part in order and write how many of each there are.

REPRODUCTIVE ANATOMY

- [ ] **Sperm, start to exit.** Trace a sperm cell from the seminiferous tubule to the outside of the body. Name every duct and every accessory gland it passes, and say what each gland adds.
- [ ] **Female internal organs.** Draw the female internal reproductive organs and label every structure. Then trace the path of an oocyte from release to the uterus.
- [ ] **Ureter, bladder, urethra.** Draw the bladder and label its wall layers and internal landmarks. Then write how the male and female urethra differ.
- [ ] **A testis in section.** Draw a testis cut open and label every structure. Then name the two cell populations inside a seminiferous tubule and what each does.
- [ ] **An ovary in section.** Draw an ovary cut open and label the follicle stages in order around the cortex, plus what is left after ovulation.
- [ ] **The perineum.** Draw the perineum and mark the two triangles. Name the boundaries and label the external structures in each sex.
- [ ] **The uterine wall.** Draw the uterus in section and label the three wall layers and the two parts of the innermost one. Then name the arteries supplying it.
- [ ] **The breast.** Draw the breast in sagittal section and label every structure inside it. Then write where its lymph drains.
- [ ] **Matched pairs.** Build a mind map pairing male and female reproductive structures that come from the same embryonic tissue. Put the shared origin in the middle.
- [ ] **The pelvic floor.** Draw the pelvic floor from above and label the muscles. Write what they support and name the openings that pass through.

RENAL, PLUS BRAIN AND MENINGES

- [ ] **Where urine is made.** Walk the nephron segment by segment and write what happens to the filtrate at each one. Name the segment before you describe it.
- [ ] **Meninges and spaces.** List the meninges from the skull inward. Name the space above or below each one and say what is in it.
- [ ] **CSF, made to absorbed.** Trace CSF from where it is made to where it is absorbed. Name every ventricle, opening, and structure it passes in order.
- [ ] **The circle of Willis.** Draw the circle of Willis as seen from the base of the brain. Name every artery in the ring and every artery joining it, and write what the ring is for.
- [ ] **Lobes and landmarks.** Draw the brain from the side and label all the lobes, the two major sulci, and the gyri on either side of the central sulcus. Write what each of those two gyri does.

- [ ] **The ventricles.** Draw the ventricular system as a single connected shape seen from the side. Name every chamber and every passage between them.
- [ ] **Dural folds and sinuses.** Draw the inside of the skull and label the dural folds. Then name the venous sinuses and trace blood from the superior sagittal sinus out of the skull.
- [ ] **Deep gray matter.** Draw a coronal slice of the brain and label the basal nuclei and the internal capsule. Name each nucleus and write what the group is broadly for.
- [ ] **Layers of protection.** Build a mind map of everything protecting the brain, working from the outside in. Put each layer on its own branch with what it protects against.
- [ ] **The diencephalon.** Draw a midsagittal brain and label the diencephalon and everything in it. Then name two limbic structures and where they sit.

BRAINSTEM AND CRANIAL NERVES

- [ ] **Twelve nerves, in order.** List all twelve cranial nerves by number and name. Beside each write sensory, motor, or both.
- [ ] **Brain, midsagittal.** Draw the brain in midsagittal view and label every structure you can. Then bracket the three regions of the brainstem.
- [ ] **The brainstem, region by region.** Name the three brainstem regions top to bottom. For each, name a structure or nucleus you can see or recall and one thing that region handles.
- [ ] **Three territories.** Draw the brain from the side and from the medial surface. Shade and label the region supplied by the anterior, middle, and posterior cerebral arteries, and write where each one comes off the circle of Willis.
- [ ] **Nerves off the base.** Draw the base of the brain and mark where each cranial nerve emerges. Name every nerve at its exit point.
- [ ] **The cerebellum.** Draw the cerebellum from behind and in section. Label every part and name what it connects to the brainstem with.
- [ ] **Thalamus and hypothalamus.** Draw the diencephalon enlarged. Mark the thalamus and the hypothalamus, label what sits in each, and write the one-line job of each.
- [ ] **Through the brainstem.** Draw the brainstem from the front and from the back. Label every surface feature and name what runs through it front to back.
- [ ] **Nerves by job.** Build a mind map grouping the cranial nerves by what they do rather than by number. Use branches for eye movement, face, throat and neck, and special senses.
- [ ] **What breaks when.** For six cranial nerves, write the nerve, one thing you would see if it stopped working, and the test that would show it.

CRANIAL NERVES, PLUS SPINAL CORD

- [ ] **Spinal cord in cross section.** Draw the spinal cord in cross section and label the horns, the columns, the roots, and the central canal. Mark which way information travels on each side.
- [ ] **Spinal nerves and plexuses.** Write the number of spinal nerve pairs by region. Then name the plexuses and one major nerve coming out of each.
- [ ] **Nerve, hole, test.** For as many cranial nerves as you can, write the nerve, the skull opening it passes through, and the one test you would use to check it.
- [ ] **Tracts up and down.** Draw the spinal cord in cross section and mark where the main ascending and descending tracts run. Name each tract and what it carries.
- [ ] **A reflex arc.** Draw a reflex arc with all five components labeled. Then write whether it is monosynaptic or polysynaptic and name a real example.
- [ ] **Dermatomes.** Draw a body outline and mark the dermatome landmarks you can remember. Name the spinal level for each.
- [ ] **Where to put the needle.** Draw the lower spinal cord in the canal. Mark where the cord ends, where the subarachnoid space ends, and the level you would enter for a lumbar puncture. Write why that level is safe.
- [ ] **Two autonomic outflows.** Draw the body and mark where the sympathetic and parasympathetic fibers leave the central nervous system. Name the ganglia each uses and write the length of the fibers.

- [ ] **Skin to brain and back.** Build a mind map following a signal from a touch on the finger all the way to a muscle contraction in the hand. Name every structure it passes.
- [ ] **The end of the cord.** Draw the lower end of the spinal cord in detail and label the conus medullaris, cauda equina, and filum terminale. Write why the roots have to travel so far.

PART C

Practice brain dump pages

Use these pages all term. One prompt per page. Retrieve first with everything closed, then check yourself in a second colour and mark what to come back to.

PRACTICE BRAIN DUMP

PROMPT ______________________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

____

____

WHAT WAS WRONG

____

____

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ______________________________

PRACTICE BRAIN DUMP

PROMPT ____________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

WHAT WAS WRONG

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ____________________

PRACTICE BRAIN DUMP

PROMPT ____________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

____

____

WHAT WAS WRONG

____

____

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ____________________

PRACTICE BRAIN DUMP

PROMPT ______________________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING	WHAT WAS WRONG

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ______________________________

PRACTICE BRAIN DUMP

PROMPT ______________________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

WHAT WAS WRONG

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ______________________________

PRACTICE BRAIN DUMP

PROMPT ______________________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

____

____

WHAT WAS WRONG

____

____

RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

REVISIT ON ______________________________

PRACTICE BRAIN DUMP

PROMPT ______________________________

EVERYTHING CLOSED. WHAT CAN I RETRIEVE?

CHECK. SECOND COLOUR.

WHAT WAS MISSING

____

____

WHAT WAS WRONG

____

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RETRIEVAL RESULT ☐ Strong ☐ Mostly there ☐ Patchy ☐ Needs work

FAILURE TYPE ☐ Could not recall ☐ Recalled it wrong ☐ Knew it, could not apply

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